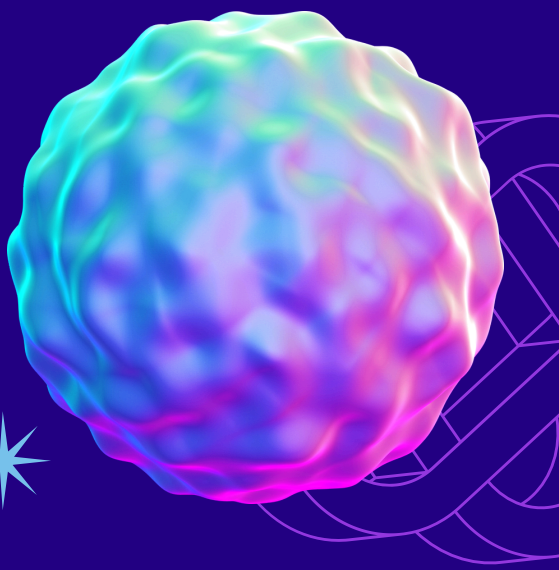
T. JOHN INSTITUTE OF TECHNOLOGY

PROCEEDINGS OF

# MULTIDISCIPLINARY NATIONAL CONFERENCE ON ADVANCES IN SCIENCE, TECHNOLOGY, ENGINEERING, AND MANAGEMENT

## NCASTEM - 2026



EDITORS:


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**Dr. Asha S Manek**

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Department of Artificial Intelligence and Machine Learning & Information Science and Engineering




Gottigere Post, Bannerghatta Road, Bengaluru -560083

Multidisciplinary National Conference on
Advances in Science, Technology, Engineering and Management
**(NCASTEM-2026)**

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The Multidisciplinary National Conference on Advances in Science, Technology, Engineering and Management **(NCASTEM-2026)** is a prestigious gathering that provides a platform for researchers, academicians, professionals, and students to share their insights, discoveries, and innovations across diverse disciplines and discuss the latest developments. The conference serves as a catalyst for fostering collaboration, promoting interdisciplinary research, and addressing contemporary challenges in **S**cience, **T**echnology, **E**ngineering and **M**anagement. It promotes a dynamic setting for knowledge sharing, creativity, and the growth of excellence in practice and research in a number of disciplines. Advancing the frontiers of **S**cience, **T**echnology, **E**ngineering and **M**anagement is crucial for the advancement of society.

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## MESSAGE FROM CHIEF PATRON

I am extremely delighted to know that the **Department of Artificial Intelligence and Machine Learning** & the **Department of Information Science and Engineering** of T. John Institute of Technology are jointly organizing the **Multidisciplinary National Conference on Advances in Science, Technology, Engineering, and Management (NCASTEM-2026)** on **30th October 2026**.

This conference serves as a premier platform bringing together researchers, academics, and scholars from across diverse domains to foster interdisciplinary collaboration and meaningful exchange of ideas. It is an exceptional opportunity to unite the research fraternity in pursuit of scientific discovery and technological innovation.

I extend my heartfelt congratulations to the organizing committee, under the leadership of **Dr. H. P. Srinivasa**, Principal, TJIT, and **Dr. Asha S. Manek**, Head of the Department of AIML, along with the faculty and team members, for their vision, dedication, and exemplary teamwork in executing this event. Their meticulous planning has laid the foundation for a highly successful and impactful conference.

As we gather to share ideas and insights, I am confident that NCASTEM-2026 will inspire new avenues of research and propel emerging technological advancements. I wish all the participants, delegates, and organizers a rewarding and memorable conference.

**Sd/-**
**Dr. Thomas John**




**Dr. H P Srinivasa**
Principal
T. John Institute of Technology, Bengaluru

## MESSAGE FROM PATRON

I am delighted to learn that the **Department of Artificial Intelligence and Machine Learning** and the **Department of Information Science and Engineering** of T. John Institute of Technology are jointly organizing the Multidisciplinary National Conference on Advances in Science, Technology, Engineering, and Management (NCASTEM-2026).

In today's rapidly evolving world, human creativity and the relentless drive for innovation continue to push the boundaries of knowledge, shaping a better future for society across diverse domains. This conference has become a significant platform for fostering innovation, collaboration and knowledge exchange in the rapidly evolving multidisciplinary fields. As we gather on our campus in the IT city of Bengaluru, this conference serves as a pivotal platform for exploring cutting-edge research and practical applications that address the complex challenges of an interconnected world.

I am confident that the deliberations, discussions, and presentations during the conference will pave the way for ground-breaking research solutions and strengthen the global research community. I extend my sincere appreciation to the organizing committee, distinguished keynote speakers, and participants for their exemplary dedication. May **NCASTEM 2026** be an enriching, impactful, and intellectually inspiring conference for all.

**Sd/-**
**Dr. H P Srinivasa**



**Dr. Asha S Manek**
Professor & HoD (AIML)
T. John Institute of Technology, Bengaluru

## MESSAGE FROM PROGRAM CHAIR

On behalf of the Program Committee, I am delighted to share the official proceedings of the **Multidisciplinary National Conference on Advances in Science, Technology, Engineering, and Management (NCASTEM-2026)**, jointly hosted by the **Department of Artificial Intelligence and Machine Learning** and the **Department of Information Science and Engineering** on **30th October 2026**, in partnership with the **International Journal of Computer Applications (IJCA)**.

In an era defined by rapid technological transformation, multidisciplinary research is essential for addressing complex real-world challenges. NCASTEM-2026 was conceptualized to bridge domain boundaries, showcase groundbreaking research, and foster dynamic knowledge exchange among scholars, industry experts, and students.

This volume brings together high-quality, peer-reviewed research contributions published in collaboration with IJCA. I extend my heartiest appreciation to all contributing authors for sharing their scholarly work, as well as to our esteemed reviewers and editorial partners for maintaining the highest academic standards. I am also deeply grateful to the leadership at TJIT and the joint department organizing teams for their unwavering commitment to this event.

I trust these proceedings will serve as a valuable reference and inspire continued research and innovation across disciplines. Thank you for your hard work and enthusiasm for growing the fields of science, technology, engineering, and management. I believe that by working together, we will achieve important progress and create new ideas that will help shape the future.

**Sd/-**
**Dr. Asha S Manek**



**Dr. Doddegowda B J**
Associate Professor (ISE)
T. John Institute of Technology, Bengaluru

## MESSAGE FROM THE CONFERENCE CHAIR

It gives me great pride to present the proceedings of the **Multidisciplinary National Conference on Advances in Science, Technology, Engineering, and Management (NCASTEM-2026)**, jointly organized by the **Department of Artificial Intelligence and Machine Learning** & the **Department of Information Science and Engineering** at **T. John Institute of Technology** on 30th October 2026.

In today's interconnected research landscape, multidisciplinary engagement is vital for driving technological breakthroughs and solving complex engineering challenges. NCASTEM-2026 has provided a dynamic national forum for scholars, researchers, industry professionals, and students to present forward-thinking ideas, exchange insights, and build meaningful academic partnerships.

We are privileged to collaborate with our esteemed publishing partner, the International Journal of Computer Applications (IJCA), to publish the high-quality research works selected through a thorough peer-review process. These proceedings represent significant efforts across diverse domains of science, engineering, and management.

I extend my deepest gratitude to our Management and Principal Dr. H. P. Srinivasa for their constant encouragement and guidance. My heartfelt appreciation goes to the Program Chair Dr. Asha S. Manek, the editorial board of IJCA, our keynote speakers, reviewers, authors, and the joint department organizing teams whose collective hard work ensured the grand success of this conference.

**Sd/-**
**Dr. Doddegowda B J**

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## ZeroTrustNet: Multi-Algorithmic Framework for Insider Threat Detection System

[1]Vaibhavi Vilas Pednekar, [2]Vedashree M R, [3]Yashaswini S, [4]Vijet Ningappa Sangam,
[5]Divya G S, [6]Sheetal S R
Dept. Computer Science and Engineering
AMC Engineering College, Bengaluru, Karnataka

**Abstract:** Traditional perimeter-based cybersecurity architectures assume that authenticated users within an enterprise bound-ary are inherently trustworthy. Consequently, insider threats originating from compromised accounts, disgruntled personnel, and careless employees remain among the most elusive attack vectors. This paper presents ZeroTrustNet, an enterprise-oriented Zero Trust security architecture designed to perform continuous post-login behavioural telemetry evaluation. Moving beyond isolated heuristic rules and single-classifier models, the proposed system introduces an explainable, six-layer hybrid intelligence engine combining deterministic policy guardrails, peer-group behavioral baselining, and unsupervised machine learning algorithms, namely Isolation Forest, Local Outlier Factor (LOF), One-Class Support Vector Machines (OC-SVM), and Density-Based Spatial Clustering of Applications with Noise (DBSCAN). The platform extracts multi-dimensional activity vectors across login temporality, bulk file exfiltration, generative artificial intelligence (GenAI) tool consumption, peripheral attachment, and contextual stress metrics. Risk is aggregated into a dynamic score ranging from 0 to 100, which triggers granular automated remediation such as step-up authentication, privilege revocation, and session quarantine. Empirical validation demonstrates that the hybrid unsupervised outlier modeling achieves a 96.4% detection rate on anomalous data exfiltration with a false-positive rate of 2.8%, providing Security Operations Center (SOC) analysts with explainable and actionable threat mitigation.

**Keywords:** *Anomaly Detection, Behavioral Biometrics, Insider Threat Detection, Machine Learning, Zero Trust Architecture, User and Entity Behavior Analytics, Explainable AI.*

## AgriCare: A Unified AI Engine for Intelligent Crop Management with Voice Assistance

[1]Asha S Manek, [2]Sylvia Tom, [3]Neha Sunil, [4]Ivin K S, [5]Abhijith R Pillai,
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru, India

**Abstract:** Agriculture is plagued by problems such as crop diseases, poor irrigation, unpredictable climate, and a lack of expert advice to farmers. Existing monitoring systems and smart farming systems are often difficult to provide useful information. This project will present a Drone-Based Precision Agriculture System using Drone LiDAR Mapping, Multi-Spectral Imaging, Computer Vision and AI to create an overall management system.

Automated drones are used to gather high quality data from above crops. AI algorithms generate Heat Maps that show how healthy the crops are, how much water is in the soil, disease risks, elevation, and irrigation issues. The system automatically detects pathogens in crops but can also provide advice to farmers on specific farms through multilingual IVR voice systems, WhatsApp, and dashboards. This system will help reduce crop waste by providing easy to understand information about the farms in real time.

**Keywords:** *Autonomous Drones, Computer Vision, Crop Disease Detection, Drone-LiDAR Mapping, Heatmap Analytics, Precision Agriculture, Smart Farming, Voice Assistance.*

## ISL Bridge - A Multimodal Indian Sign Language Recognition System for Inclusive Communication

[1]K V Shalini, [2]Sri Harsha T, [3]Bindhu Monika A K, [4]Shivalingayya Hiremath
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** ISL - Indian Sign Language is an essential part of communication for hearing and speech impaired people; but there are communication barriers between sign language users and non-users of sign language. This paper describes an AI -based Indian Sign Language Recognition and using Deep Learning Translation System is developed and Computer Vision to identify hand gestures and translate those gestures into meaningful text output. the proposed system provides both uploaded images of gestures and real-time predictions based on Webcam images of gestures. The system uses MediaPipe Hands for accurate extraction of hand landmarks. The gesture classification is performed using a TensorFlow/Keras deep learning model that is trained on the features extracted from the hand landmarks. The system is built using Flask, OpenCV, HTML, CSS, JavaScript, and the Google Translate API. The system is a responsive website that creates an accessible communication platform. Once the gesture is recognized by the system, it will also be translated into Kannada for accessibility in the regional areas where this system is implemented. The experimental results thus demonstrate that this system can perform accurate, real-time, and lightweight gesture recognition for use in assistive communication applications.

**Keywords:** *Assistive Communication System, Gesture Recognition, Indian Sign Language (ISL), Kannada Translation, MediaPipe Hands, OpenCV, TensorFlow/Keras.*

## AI-Based Phishing and Fake Website Detection System Using Machine Learning and Real-Time Threat Analysis

[1]Suhas S, [2]Snigdha Kesh
Dept. Computer Science and Engineering,
AMC Engineering College

**Abstract:** Phishing and counterfeit websites represent significant cybersecurity risks because attackers disguise links and web pages to collect confidential user information. Conventional blacklist methods can identify reported threats, but they may fail when attackers create fresh or altered malicious addresses. This paper introduces PhishSleuth, an AI-driven system that examines URL properties and uses machine-learning classification to recognize websites that may be fraudulent. After receiving a URL, the system derives structural and security indicators, prepares the resulting feature vector, and supplies it to a trained classification model. The prediction is shown as legitimate, suspicious, or phishing, and risky inputs generate a user warning. The architecture also supports future integration of real-time threat-intelligence services and browser-based protection.

**Keywords:** *Phishing detection, fake website detection, machine learning, artificial intelligence, URL analysis, cybersecurity, threat detection, real-time analysis.*

## Threatgraph: An Explainable Knowledge Graph-Based Framework for E-Commerce Fraud Detection and Cybersecurity Investigation

[1]Kanika Chalana, [2]Rakesh M, [3]Kritik Munot, [4]Anjana V
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** E-commerce fraud has become a significant challenge with the rapid growth of digital transactions and online payment systems. This paper presents ThreatGraph, an explainable Knowledge Graph-based framework for e-commerce fraud detection and cybersecurity investigation. The proposed framework represents transactions and associated entities as a temporal Knowledge Graph to identify suspicious relationships and historical fraud patterns. An explainable rule-based detection mechanism is used to generate evidence and assess transaction risk, classifying transactions as Safe, Suspicious, or Fraud. The framework further integrates a six-dimensional security assessment covering authenticity, integrity, confidentiality, access control, security posture, and non-repudiation. SecureBERT 2.0 is incorporated for cybersecurity-domain natural language processing, while ThreatFox provides external threat-intelligence enrichment for relevant indicators. The system also supports investigation case management to organize evidence and assist analysts in reviewing suspicious activities. The framework is evaluated using the IEEE-CIS Fraud Detection dataset through chronological, leakage-controlled evaluation, with the aim of assessing detection performance while preserving historical information constraints and supporting explainable investigation.

**Keywords:** *ThreatGraph, E-Commerce Fraud Detection, Knowledge Graph, Explainable Artificial Intelligence, Cybersecurity, SecureBERT 2.0, Threat Intelligence, Temporal Fraud Detection.*

## Equity-Weighted Urban Hazard Prioritization Using a Dual-YOLO Detection Framework for Road and Pedestrian Safety

[1]Khuswanti Sirvi, [2]Rupam Talukdar, [3]Jaya Karuna, [4]Dhaarna Purna Khatri, [5]Bhavana Singh
Dept. Computer Science and Engineering,
AMC Engineering College, Bengaluru, India

**Abstract:** Indian cities produce urban disamenities that can harm pedestrians, like garbage and potholes, at a rate much higher than city agencies can discover and repair. Current response to civic complaints is inefficient and also ineffective, often favouring areas with high complaint rates. This paper introduces an AI-based road inspection framework, an image-based approach that uses a street view imagery-based dual-detector pipeline-YOLOv8 (for garbage detection) and YOLOv12s (for pothole and crack detection)-to estimate and analyse the level of severity of infrastructure and pedestrian hazards in cityscapes. These detected hazards are quantified into an Urban Decay score and a Pedestrian Safety score, and these two are weighted into an overall Equity-weighted Priority score, multiplying hazard level by equity weights based on pedestrian and population densities, and closeness to areas with sensitive populations. In experiments, our garbage detector attained an overall precision of 51.4%, recall of 40.0%, and mAP@50 of 37.8% on the test dataset and self-collected small dataset, while a system-wide test of our framework on 19 segments in Bengaluru identified 15 of them to have high priority level civic defects. We show that the system- generated heat map reprioritizes maintenance toward high-vulnerability segments.

**Keywords:** *Yolov8, urban infrastructure monitoring, pedestrian safety, equity-weighted prioritization, heatmap, smart cities, computer vision.*

## SignBridge AI: Real-Time Sign Language Recognition Using MediaPipe and BiLSTM

[1]Mamtha C. R., [2]Subashree Jeyaraman, [3]Harshita Venkatesh Kundgol
Dept. Computer Science and Engineering (Data Science)
Vemana Institute of Technology, Bengaluru-India

**Abstract:** Communication becomes difficult when two people do not understand the same language. This can be a challenge for sign-language users when they interact with people who are unfamiliar with their signs. SignBridge AI is a camera-based project designed to recognize a limited set of hand gestures and present the predicted meaning as text and speech. The system uses a webcam to capture hand movements and MediaPipe Hands to locate 21 landmarks on the hand. Each landmark contains three coordinate values, producing 63 features for every frame. These features are normalized relative to the wrist and collected over 30 consecutive frames. A bidirectional long short-term memory (BiLSTM) model processes the sequence and predicts the gesture class. The predicted result is then displayed through a user interface and can be spoken using a text-to-speech component. The project uses Python, PyTorch, MediaPipe, FastAPI, and a React/Vite frontend. This paper explains the problem, system design, implementation, possible evaluation methods, limitations, and future improvements. The supplied project report does not contain a verified test accuracy, complete dataset details, or measured end-to-end response time. Therefore, no unverified performance figures are presented as experimental results.

**Keywords:** *Sign language recognition, MediaPipe, BiLSTM, hand landmarks, gesture classification, computer vision, text-to-speech.*

## Deepshield-X : Deepfake Detection System

[1]Madhavi Dixit, [2]Poojitha K, [3]Chandana K Gowda, [4]Chinmayi Kule, [5]Nagakalyan V
Department of Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** With advances in Generative AI, there is a surge of Deepfake media. Unimodal detection models are problematic at detecting different layers of manipulation. In this paper we present Deep shield-X, a multimodal detection model. This detection system uses a combination of Convolutional Neural Networks, Transformers and Temporal Consistency Analysis to detect pixel manipulation, audio anomalies, unnatural movement and audio-visual misalignment. The model uses a fusion layer to produce an overall score based on the decisions made for each modality. The model also provides explainable AI heatmaps. The model has been implemented as a serverless full stack application using React.js. It has a high detection rate of 90-95% and can process an image in between 1 and 3 seconds. This demonstrates a highly accurate detection system by using the fusion of multiple modalities through cloud-based infrastructure.

**Keywords:** *AI Deepfake Detection, Multimodal Deep Learning, Convolutional Neural Network (CNN), Transformer Models, Explainable AI (XAI), Digital Forensics, Media Authentication, Audio-Visual Analysis.*

## ASSET TOKENIZATION: Decentralized Platform for Real-World Asset Tokenization Using Blockchain

[1]Omkar Bharath Pukale, [2]Premkumar M K, [3]V Mareeswari, [4]Sahil Kumar Mishra,
[5]Sandesh G C, [6]Sanjitha A
Dept. Computer Science and Engineering
AMC Engineering College, Bengaluru, Karnataka

**Abstract:** Asset tokenization enables real-world assets to be represented as fractional digital tokens on a blockchain, allowing ownership to be divided into smaller units. This paper presents a web-based asset tokenization platform that integrates React.js, Node.js, Express.js, MongoDB, Solidity smart contracts, and MetaMask. The system uses ERC-20 tokens to represent fractional asset ownership and supports asset listing, token creation, token purchase, transfer, and ownership verification. Smart contracts are developed and tested using Hardhat and deployed on the Ethereum Sepolia test network for evaluation. The prototype demonstrates the integration of conventional web technologies with blockchain-based ownership management and provides a transparent mechanism for recording token transactions. The system is evaluated through functional, smart-contract, and integration testing to validate the proposed workflow.

**Keywords:** *Asset Tokenization, Fractional Ownership, Blockchain Technology, Smart Contracts, Ethereum, Sepolia, ERC-20.*

## AI-Based Personalized Internship Recommendation System

[1]Madhavi Dixit, [2]Abdul Hameed, [3]Abdul Rahman, [4]Subhaib Dar, [5]Mirza Hamza Baig
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Getting a good fit intern for a suitable internship program is difficult for students because there are significant differences between internships concerning technical skills, educational background, job roles, preferences, and career interest. Current internship systems use keyword-based search techniques along with manual input of student preferences that might be unable to properly describe behavior and hidden interests of students. This paper suggests developing a personalized AI-based Internship Recommendation System that uses various aspects of student's profile such as technical skills, educational background, interests, career preferences, and behavioral responses. In particular, the system will have a behavioral assessment feature that will use scenario-based questions to reveal preferences, working style, problem-solving approach and career interests of students. All this information will be used together with requirements for an internship program to provide personalized recommendations. The suggested model includes student profiling, behavioral analysis, feature extraction, internship data processing, similarity calculation, recommendation generation, and improvement through feedback. The goal of the system is to filter out irrelevant recommendations and help students find internships that would suit them more.

**Keywords:** *Behavioral Analysis, Career Recommendation, Internship Recommendation, Machine Learning, Natural Language Processing, Personalized Recommendation, Recommender System, Student Profiling.*

## Real-Time Multilingual Sign Language Translation Using Hierarchical CNN and Geometric Heuristics

[1]Rounit Kashyap, [2]V. R Srividhya
Dept. Computer Science and Engineering
RVITM, Bengaluru, India

**Abstract:** Sign language gives a communication platform to those who are deaf or have hearing problems, but such people face difficulties communicating with those who do not know sign language. The research described in this paper introduces a real-time multilingual recognition and translation system, which utilizes hand landmark abstraction, convolutional neural networks, and geometric heuristics. The system extracts 21 three-dimensional hand landmarks from a webcam using MediaPipe Hands, which is then normalized to appear on a 400 × 400 white canvas to minimize background, color, and lighting distractions. The article describes a custom-built dataset containing 26 alphabetical classes with approximately 180 samples each (4,681 images in total). The research classifies each character using landmarks through a hierarchical system, which classifies them based on gestures fall into eight visually similar groups. It uses geometric heuristics to classify characters by applying scale-invariant landmark distance ratios. The system constructs sentences from recognized characters and translates them into Hindi and Kannada using a dictionary-based approach. Moreover, the system provides transliteration and text-to-speech synthesis in multiple languages, facilitating communication. The findings reveal that the system attains up to 99% accuracy in character classification under different environmental conditions.

**Keywords:** *Sign Language Recognition, MediaPipe, Convolutional Neural Network, CNN, Hand Landmark Detection, Geometric Heuristics, Multilingual Translation, Text-to-Speech, Computer Vision, Assistive Technology.*

## Statistical Study of Population Growth

[1]J. Raji, [2]Shashidhar S, [3]Pavan D S, [4]V. Yatin, [5]Akash L B, [6]Akshay Gowda
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Collection of global population figures often hide significant regional differences. This paper presents a quantitative analysis of global and regional population dynamics from 2010 to 2026, using empirical demographic data (Census) compiled from the United Nations Population Division (2024/2026 Revisions). We evaluate compound growth rates, linear expansion trends, and non-linear demographic transition patterns, and we use structural proportion metrics and growth contribution indices to compare six continental divisions: Asia, Africa, Europe, Latin America and the Caribbean, Northern America, and Oceania. The results show that the world population reached a historic peak of approximately 8.32 billion in 2026, while the relative growth rate continues to slow. Growth is heavily concentrated in Africa and Asia, whereas Europe and Northern America show stabilizing populations and structural demographic aging. Taken together, the findings indicate a transition from exponential expansion to a near-linear trajectory, with the regional distribution of the world's people shifting steadily toward Africa.

**Keywords:** *Global population, Regional demographics, CAGR (Compound Annual Growth Rate), Demographic Transition, Growth Contribution Index, Population Projection.*

## Counterfeit Medicine Detection: A Comprehensive Review of Analytical, AI-Based, And Digital Authentication Methods

[1]Shivasharanappa S A, [2] Sudeep S M, [3] Vinay S, [4] Yeshwanth Gowda K S
Dept. Electronics and Communication Engineering
AMC Engineering College

**Abstract:** Substandard and falsified medicines remain a serious public-health concern because they can contain incorrect ingredients or doses, fail to deliver the expected therapeutic effect, or expose patients to harmful contaminants. Detection is challenging: packaging and digital identifiers can be copied, while laboratory methods that examine chemical composition may require costly instruments and trained operators. This review critically surveys complementary approaches to medicine authentication, including visual inspection and computer vision, machine learning, near-infrared (NIR) and Raman spectroscopy, other analytical methods, Internet of Things (IoT) sensing, and blockchain-enabled track-and-trace systems. Particular attention is given to low-cost edge-AI systems that combine tablet appearance and physical measurements. The supplied project report describes a Raspberry Pi-based prototype using a camera, OpenCV-derived tablet features, manually entered mass, and a Random Forest classifier to assign samples to genuine, substandard, or counterfeit classes. Such a system may support rapid preliminary screening, but appearance, mass, and diameter alone cannot establish chemical identity, active pharmaceutical ingredient content, dissolution performance, or safety. The review therefore distinguishes screening from confirmatory testing and proposes a validation framework based on representative sampling, batch-level separation, sensitivity, specificity, calibration, external testing, and uncertainty reporting. A layered strategy combining supply-chain traceability, physical/visual screening, portable spectroscopy, and laboratory confirmation is identified as a more defensible direction than reliance on any single method.

**Keywords:** *Counterfeit medicines, falsified medicines, substandard medicines, computer vision, machine learning, Random Forest, spectroscopy, Raman, near-infrared, blockchain, pharmaceutical quality.*

## Diabetic Foot Ulcer Severity Classification Using MobileNetV2: A Transfer Learning Based Approach

[1]S Nethravathi, [2]Disha Prakasha, [3]Shamaim Huda, [4]Rithika K, [5]Tejas G
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Diabetic foot ulcers (DFUs) are a common and dangerous complication of long-term diabetes, and catching them early can prevent serious infections and, in severe cases, amputation. This paper presents Phase 1 of an ongoing project that uses a lightweight deep learning model, MobileNetV2, to automatically classify foot ulcer images into four severity grades. Transfer learning was used to fine-tune MobileNetV2 on a foot ulcer image dataset of about 10,000 images, and the model was trained and tested under four different train-test split ratios - 75:25, 80:20, 85:15, and 90:10 - to study how the amount of training data affects performance. The 85:15 split gave the best result, reaching a validation accuracy of 89.27% with a weighted F1-score of 0.89. The paper also reviews related deep learning approaches reported in recent literature and discusses the strengths and current limitations of the MobileNetV2-based model, along with the next steps planned for later phases of the project.

**Keywords:** *Convolutional Neural Network, Diabetic Foot Ulcer, MobileNetV2, Transfer Learning, Severity Classification, Medical Image Analysis*

## Arduino and Vision-Based Blind Spot Detection System Using Ultrasonic Sensors and OpenCV

[1]Goutam G Godakhindi, [2]Nandita Bangera
Dept. Computer Science and Engineering
RV Institute of Technology and Management (RVITM), Bengaluru, India

**Abstract:** Blind spots are areas surrounding a vehicle which cannot be properly seen by the driver using the standard mirrors. Objects situated in these areas might fail to be noticed when changing lanes, passing another vehicle, parking or turning, thus raising the chance of a collision. Traditional Blind Spot Detection (BSD) systems usually make use of radar, ultrasonic sensors, cameras, or a combination of these technologies. Even though radar-based systems designed for use in cars offer long-range detection, their high cost and implementation complexity can make them inappropriate for low-cost educational and prototype uses. This paper describes a low-cost hybrid blind spot detection prototype that combines an Arduino microcontroller, ultrasonic sensors, a camera, Python, and OpenCV. The ultrasonic sensing section monitors the left and right sides of the vehicle and sends its detection results to a computer via serial communication. The computer-vision section processes the camera frames and detects objects such as cars, bicycles, pedestrians, and other pertinent objects. If an object is found in a monitored blind-spot area, the system issues a directional warning to the driver, with a buzzer and an LED providing an immediate local alert. The proposed design separates short-range proximity detection from visual object recognition, enabling the two methods to reinforce one another. The prototype shows that it is possible to build an affordable educational blind-spot detection platform using readily available hardware and open-source software, and it lays the groundwork for future enhancements such as machine-learning-based detection, object tracking, sensor fusion, wireless communication, and embedded deployment.

**Keywords:** *Embedded Systems, Computer Vision, Vehicle Safety, ADAS, Arduino, Blind Spot Detection, Object Detection, OpenCV, Python, Ultrasonic Sensor.*

## AI-Enabled Wearable Cardiac Monitoring and Integrated Emergency Healthcare Coordination

[1]S Nethravathi, [2]Prarthana P, [3]Anakha Raj, [4]Nithya Rekha R K
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Time to diagnosis in emergencies is an important factor to consider when providing healthcare. In order to provide quick and accurate diagnoses and ensure smooth communication between clinics, emergency services and hospitals patients will each be given a unique ID which stores their medical history. Patients will then wear a band with embedded sensors which take ECG, heart rate and GPS readings. The band also contains an AI module which will monitor the heart rate in real time. This will detect any dangerous irregularities. Signal analysis is used to avoid the AI falsely activating. The system will automatically dispatch an ambulance. It will calculate the fastest route to the closest ambulance based on traffic. The system will also determine the most suitable hospital for the patient based on their equipment. The patients' vitals and medical history will then be streamed to the hospital so they can give priority treatment.

**Keywords:** *Artificial Intelligence, Ambulance Coordination, Cardiac Monitoring, ECG Analysis, Emergency Healthcare, GPS Tracking, Hospital Coordination, Wearable Sensors*

## Machine Learning-Based DDoS Detection and Prevention System with Real-Time Monitoring Dashboard

[1]S. Saila, [2]Saniya Sankhya Hegde, [3]Yallutla Mounika, [4]Shrilakshmi T. Hegde
Dept. Computer Science and Engineering
Jyothy Institute of Technology, Tataguni, Bangalore, India

**Abstract:** Distributed Denial-of-Service (DDoS) attacks are still one of the most damaging threats facing networked systems today, and signature- or threshold-based defences often can't keep up with new attack patterns and sudden spikes in traffic. This paper presents DDoS Guard, a full-stack DDoS detection and prevention system built around machine learning, with a real-time monitoring dashboard for analysts. The core of the system is a three-layer detection pipeline: a Random Forest classifier trained on the CICIDS2017 dataset, an Isolation Forest that scores how unusual incoming traffic looks, and a heuristic rule engine that catches obvious flood patterns. These three outputs are combined into a single confidence score, and when an attack is confirmed, a mitigation engine takes over, applying token-bucket rate limiting and temporarily blocklisting the offending IP addresses. Everything happening on the network is streamed live to a React dashboard over Server-Sent Events (SSE), so analysts can watch traffic, alerts and forensic reports as they happen. JWT authentication and role-based access control keep administrator and analyst permissions separate, and administrators are notified automatically by email whenever a high- or critical-severity incident is logged. This paper walks through the system architecture, the mathematics behind the ensemble decision and the mitigation logic, and the evaluation approach used to measure detection accuracy and response time, before discussing where the current system falls short and how it could be extended.

**Keywords:** *Algorithms, Design, Experimentation, Measurement, Performance, Security, DDoS Detection, Machine Learning, Random Forest, Isolation Forest, Real-Time Monitoring, Network Security, Intrusion Prevention, Ensemble Learning.*

## Integrated Multi-Modal System for Real-Time Deepfake Video Detection

[1] K P Agarwal, [2]Abhay Aditya R S, [3]Naveen Kumar H B, [4]Abhishek Sharma, [5]Ishaq Ahmed Patel
Dept. of Information Science and Engineering
T. John Institute of Technology, Bengaluru-83

**Abstract:** The rapid development of synthetic media has made it increasingly difficult to distinguish authentic videos from manipulated content. This paper presents a lightweight web-based Deepfake Video Detection System that analyses uploaded videos for visual and temporal inconsistencies. The proposed system uses a Flask-based backend and OpenCV for video processing, frame sampling, and face detection. Up to 24 frames are sampled from the input video and examined for compression artifacts, temporal flicker, face motion instability, and color and brightness inconsistencies. These signals are combined using a weighted heuristic scoring mechanism to estimate whether a video appears REAL, FAKE, or UNCERTAIN. The browser-based interface displays the prediction, confidence level, suspicious signals, and analysis metrics. The proposed implementation serves as a lightweight proof of concept for preliminary deepfake screening rather than forensic-grade verification.

**Keywords:** *Deepfake Video Detection, Synthetic Media, Video Forensics, OpenCV, Flask-Based Web Application, Facial Motion Analysis, Temporal Inconsistency Detection, Heuristic Scoring, Video Authenticity Verification, Explainable AI.*

## Trip Companion: Tour Together and Expense Tracker Application

[1]Madhavi Dixit, [2]Druva Prasad K L, [3]K Dilli Kumar Reddy, [4]Barath R
Dept Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Group travel is increasingly popular, yet the process of organizing shared trips and settling shared costs remains fragmented across chat applications, spreadsheets, and separate expense-splitting tools. This paper presents Trip Companion, a web-based application that unifies collaborative tour planning with automated expense tracking in a single platform. Registered users can post tours describing destination, itinerary, and travel interests, enabling other travelers to discover and join trips that match their preferences. Once a group is formed, members can record individual expenses, and the system automatically computes each participant's fair share, generating transparent, real-time expense reports that eliminate manual calculation and reduce disputes. For safety during travel, the system also provides secure, role-based access and an optional, consent-based live GPS location sharing feature that lets group members and family members track a traveler's location. The application is implemented using a PHP 8.1 and MySQL back end, an HTML5, CSS3 and Bootstrap front end, and JavaScript with jQuery for dynamic client-side interaction, with a complementary Android client built in Android Studio. Functional testing confirms that the system reliably supports tour creation, member coordination, and accurate expense splitting across varying group sizes, offering a practical, centralized alternative to the disjointed tools travelers currently rely on.

**Keywords:** *Design, Algorithms, Human Factors, Management, Group Travel, Expense Tracking, Collaborative Trip Planning, Web Application, GPS Location Sharing, Travel Safety, PHP, MySQL, JavaScript, Android Studio, Cost Splitting.*

## Flexi-Fit: Scalable Gym Trainer Subscription Model

[1]P. Shoaib Aqeel, [2]Geena Gorge, [3]V. Mareeswari, Sandesh K. D, [5]Prajwal, [6]Veeresh H
Dept. Computer Science and Engineering
AMC Engineering College, Bengaluru, Karnataka

**Abstract:** Fitness users often need training services that can accommodate different schedules, budgets, and goals. A conventional gym membership may tie a user to one facility or trainer and can therefore be inconvenient when those requirements change. Flexi-Fit is proposed as a subscription platform that brings users and certified fitness trainers together in one system. Users can review trainer specializations and subscription costs, choose a suitable trainer, and change trainers when required. The platform supports online as well as in-person training, session booking, progress tracking, and communication. Cloud services and mobile-oriented technologies provide the infrastructure for these functions. Trainers can maintain their profiles, define subscription plans, organize schedules, and respond to client requests. The resulting platform combines trainer selection, subscriptions, scheduling, payments, and progress management in a single fitness service.

**Keywords:** *System, Subscription Model, Cloud Computing, Fitness Platform, System Architecture, Modern Web Technologies.*

## The Role of Artificial Intelligence in Shaping Human Life

[1]J. Pallavi, [2]Likitha S., [3]Pooja U., [4]Sparsha M.
Dept. Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Artificial Intelligence is growing more prevalent in our everyday lives, and it will continue to play a bigger role throughout 2025–2026. AI can help people in many different areas like academics, medical, communication, technology, and even at work. AI enables tasks to be completed more efficiently, improves decision-making and generates new approaches to problem-solving. At the same time, however, it also presents challenges such as job losses, concerns about privacy, and the requirement for new skills. This abstract examines the way AI is changing human life, the benefits it offers, and the responsibilities that must be followed in its use. The aim is to demonstrate how AI can be used in a safer, easier, and more positive way.

**Keywords:** *Artificial Intelligence Effect, Tools of the Future, Human Life Changes, Benefits and Future Challenges.*

## Digital Slot Booking and OTP-Based Verification System for Government-Issued BPL Cardholders

[1]Priya I. Nalavadi, [2]Pruthvi GN, [3]Pruthvi R, [4]Ramya G, [5]Soujanya C K
Department of Information Science and Engineering, T. John Institute of Technology

**Abstract:** The suggested Smart Ration Management System improves the Public Distribution System (PDS) for BPL cardholders by decreasing waiting time and crowd. It gives time-slot reservation, allowing eligible candidates to check available slots and book a suitable time for ration collection. The system uses Python Flask, MySQL, HTML, CSS, and JavaScript for secure and systematic management. Mobile-number-based OTP authentication provides secure login for individuals. Users can view ration-card details, check monthly allocation, stock availability, and registered ration shops. Digital tokens and biometric verification help verify individuals and maintain exact distribution records.

**Keywords:** *E-Governance, Digital Public Services, Information Management, Web-Based System, Database Management, BPL Card, Public Distribution System (PDS), Smart Ration Management, Time-Slot Reservation, OTP Authentication, Digital Token, Python Flask, MySQL.*

## AI-Driven IoT-Based Smart Water Quality Monitoring and Drinking Water Safety Prediction System

[1]Preethi J, [2]Shwetha K R, [3]S R Sheetal, [4]Sathya S, [5]Riva Cardoza, [6]Pooja Aski
Dept. Computer Science and Engineering
AMC Engineering College, Bengaluru, Karnataka, India

**Abstract:** This work presents an IoT enabled setup for observing drinking water quality using an ESP32 with pH, turbidity and Total Dissolved Solids (TDS) sensing. GPS records the test location and an LCD shows the measurements at the device. The readings and location information are sent through Wi-Fi to ThingSpeak for storage and remote viewing. A machine learning stage estimates chlorine concentration and a Drinking Water Safety Score without using a dedicated chlorine sensor. Green and red LEDs provide a quick local indication based on the selected limits. By combining measurement, location tagging, cloud storage and prediction, the system provides a compact framework for repeated field observations and reduces the need for manual recording.

**Keywords:** *Water Quality Monitoring, Internet of Things (IoT), Machine Learning, ESP32, pH Sensor, Turbidity Sensor, Total Dissolved Solids (TDS), ThingSpeak, GPS, Chlorine Concentration Prediction.*

## A Comparative Study of Probability Distributions and Their Applications in Real-World Data Analysis

[1]J Raji, [2]Vinay Kumar Maurya, [3]Raghu J H, [4]Yeshas L.P, [5]Rajat Mahapatra
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Probability distributions provide mathematical models for describing random variation in measurements, counts, waiting times, and bounded quantities. This paper compares five commonly used distributions: Normal, Binomial, Poisson, Exponential, and Uniform. The study reviews supplied literature on statistical distributions, parameter estimation, bounded models, and uncertainty, then compares the selected distributions by support, parameters, moments, assumptions, applications, and limitations. Five illustrative numerical examples demonstrate how probability calculations are performed for examination marks, product inspection, service requests, waiting time, and values on a fixed interval. These examples use assumed parameters and are not results from collected datasets or team-run experiments. The comparison shows that distribution selection depends on the variable type, support, data behaviour, and assumptions of the application; no single distribution is appropriate for every dataset.

**Keywords:** *Probability distributions; Normal distribution; Binomial distribution; Poisson distribution; Exponential distribution; Uniform distribution; statistical modelling; real-world data analysis.*

## Home Alone: An AI-Powered, Low-Cost Smart Home Security System Using Computer Vision, Edge Computing and n8n Workflow Automation

[1]Nethravathi S, [2]Beeresh Gowda, [3]Balaji N, [4]Karanam Rithvik Sai, [5]Yugaa Soorya Prabha A M
Dept. Artificial Intelligence and Machine Learning,
T. John Institute of Technology, Bengaluru-83

**Abstract:** Conventional home surveillance cameras record footage but do not understand it, so a homeowner must watch the feed or review recordings after an incident has already occurred. More capable security systems add motion sensors, smart locks and cloud services, which raises cost and installation effort beyond what many households can afford. This paper presents Home Alone, a software-first smart home security system that turns an ordinary camera into an automated security assistant. The system captures frames from an existing webcam, smartphone camera or Raspberry Pi camera, detects people using a pre-trained computer-vision model, and evaluates each detection against a configurable security rule set that covers restricted zones, armed hours and dwell time. When a rule is violated, an event is sent by webhook to an n8n workflow that saves a snapshot as evidence, writes a timestamped event record, sends an alert to the owner and updates a web dashboard. A Raspberry Pi serves as an optional low-cost edge device, so the prototype can be built with no additional hardware spending using resources available in a college. The paper describes the proposed methodology, system architecture, implementation approach and an evaluation plan, and discusses its advantages, limitations and future extensions such as face recognition, smoke and fire detection and IoT sensor integration.

**Keywords:** *Computer Vision, Edge Computing, Home Security, n8n Workflow Automation, Smart Home Surveillance, Threat Detection, Video Analytics*

## DNA Sequence-Based Disease Prediction Using Deep Learning: A Survey

[1]Ajay, [2]Suhani Keni, [3]Talanki Varshini, [4]Vijaya Nikhilesh, [5]Rajashree Ingale
Dept. Artificial Intelligence and Data Science
K.S. School of Engineering and Management, Bengaluru, India

**Abstract:** The increasing availability of genomic and single-nucleotide polymorphism (SNP) information has made computational analysis a useful direction for studying disease-associated patterns. At the same time, genomic records are high-dimensional, heterogeneous, and often unevenly distributed across disease classes. This survey examines learning strategies used for DNA- and SNP-oriented disease analysis, with attention to sequence representation, preprocessing, feature reduction, convolutional and recurrent architectures, synthetic augmentation, transformer models, graph learning, explainability, privacy, and scalable implementation. The review uses the forty references supplied in the project Phase-I report. The surveyed studies show that deep networks can learn useful representations from genomic inputs, while hybrid architectures attempt to capture local sequence structure and broader dependencies. Important issues include class imbalance, data quality, generalization, computational requirements, privacy, and model transparency. Feature selection can simplify the input space, but it must be performed without allowing test information to influence model development. Explainable AI methods such as SHAP can describe model attribution, although an attribution should not be treated as biological proof. Overall, the literature supports integrated disease-prediction systems that combine reliable preprocessing, efficient representation learning, transparent evaluation, and responsible handling of genomic information.

**Keywords:** *DNA sequence, SNP, disease prediction, deep learning, CNN, feature selection, explainable AI, genomic data, multi-disease classification.*

## Hand Gesture Recognition for Speech Impaired Using American Sign Language and Convolutional Neural Networks

[1]Manjunatha Swamy C, [2]Aarush Kumar
Department of ADSE
Christ University, Bangalore, India

**Abstract:** To improve Human-Computer Interaction (HCI) for individuals with disabilities, such as hearing or speaking impairments and stroke patients, through the development of a hand gesture recognition system. The system employs image processing techniques and AI classifiers, including VGG-16 (Visual Geometry Group) and Convolutional Neural Network (CNN), to accurately extract hand gesture features using American Sign Language (ASL) hand signs. Performance evaluation is conducted to assess various algorithms and determine the most effective method for real-time hand gesture recognition. The proposed system's goal is to provide a more accurate and efficient method for individuals with disabilities to communicate through hand gestures, enhancing their quality of life. The proposed CNN-based system attained the highest accuracy of 97.5% compared to existing models, as the SVM model achieved an accuracy of 89.3% using handcrafted features and the RNN model achieved 92% by capturing temporal dependencies in gesture sequences. This approach can be integrated into various applications, including virtual assistants, gaming, and healthcare, making it a valuable contribution to the field of HCI (Human-Computer Interaction) and assistive technology.

**Keywords:** *Hand Gesture Recognition, VGG-16, CNN, American Sign Language, Human-Computer Interaction, Support Vector Machine.*

**ARGUS: Agentic Resilient Graph-Based Understanding System**

[1]Abhishek B C, [2]P Akash, [3]Aryakantha K S, [4]Anish Bhande,[5]Vimuktha E. Salis
Department of Information Science and Engineering
Global Academy of Technology, Bengaluru, Karnataka, India

**Abstract:** Legacy maritime logistics systems are reactive by design, and that has become a real liability as geopolitical shocks and climate-driven disruptions grow more frequent and harder to predict. Most of these systems simply were not built to reason about how a disruption in one region ripples across a global trade network. This paper presents ARGUS (Agentic Resilient Graph-based Understanding System), an autonomous framework that pairs Graph Retrieval-Augmented Generation (GraphRAG) with Reinforcement Learning (RL) to sense disruptions as they emerge and reroute maritime shipments before delays compound. ARGUS draws on live event streams from the GDELT database, port logs, and weather feeds, folding them into a continuously updated Knowledge Graph of trade dependencies. When a disruption is flagged, an RL-driven Decision Agent simulates alternative routes and selects the one that best balances latency, cost, and carbon impact, treating sustainability as part of the optimization itself rather than a constraint checked afterward. The resulting recovery plans, along with supporting analytics, are surfaced through an interactive monitoring dashboard so that supply chain managers can see not just what ARGUS recommends, but why. Taken together, this approach is meant to move maritime disruption management from a manual, reactive process toward something closer to autonomous, explainable, self-healing infrastructure suitable for real-world logistics operations.

**Keywords:** *Agentic AI, GraphRAG, Reinforcement Learning, Supply Chain Resilience, Maritime Logistics, Knowledge Graph, Carbon-Aware Routing, GDELT.*

**Application of Modular Arithmetic for Secure Digital Authentication and OTP Generation**

[1]J. Raji, [2]Shalini A, [3]Shravya V, [4]Zeeshan Karrur, [5]Sreesha
Dept, Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Digital authentication plays a vital role in protecting user accounts, confidential information, and online transactions from unauthorized access. However, conventional authentication methods remain vulnerable to password theft, replay attacks, and interception of one-time passwords (OTPs). This paper proposes a modular-arithmetic-based approach for secure digital authentication and OTP generation. The proposed approach explores the use of modular operations, pseudorandom number generation, cryptographic hashing, and time-dependent verification to generate and validate OTPs securely. By incorporating a shared secret and a time-based counter, the system aims to generate short-lived authentication codes that reduce the risk of OTP reuse and unauthorized access. The study examines the relevance of mathematical operations, including modular reduction and modular exponentiation, to the design of authentication mechanisms and evaluates the proposed approach in terms of security, computational efficiency, verification time, and resistance to common authentication attacks. Unlike relying on modular arithmetic alone, the approach considers its integration with established cryptographic primitives to improve the reliability of digital authentication. The study highlights the importance of combining mathematical foundations with established cryptographic techniques to develop reliable and efficient authentication systems.

**Keywords:** *Digital Authentication, Information Security, Number Theory, Identity Verification, Authentication Protocols, Multifactor Authentication, Cybersecurity, Modular Arithmetic, One-Time Password Generation, Secure Digital Authentication, Cryptographic Algorithms, Time-Based One-Time Password (TOTP), Replay Attack Prevention, Computational Efficiency, Secure Verification, Digital Identity.*

## Trust Circle AI: AI-Powered Trusted Social Activity Platform Connecting Verified People

[1]Vineeta, [2]Aditya Chauhan, [3]Daksh Saini, [4]Devansh Pundir, [5]Kirti Singh
Department of CSE-AI
ABES Institute of Technology, Ghaziabad, Uttar Pradesh, India

**Abstract:** Finding trusted companions for real-world activities such as sports, study sessions, travel, events, and social gatherings can be challenging when users have limited information about the people they meet through online platforms. Existing social networking platforms primarily focus on friendship, communication, or event discovery and may not provide sufficient mechanisms for identity verification, intelligent activity-based matching, and continuous trust evaluation. This paper presents Trust Circle AI, an AI-powered trusted social activity platform designed to help users discover and connect with verified individuals for real-life activities. The proposed platform combines identity verification, AI-based smart matching, Dynamic Trust Score, and AI-based safety monitoring to create a more trusted environment for real-world interactions. The system considers factors such as interests, location, preferences, activity type, participation history, ratings, and community feedback to recommend compatible companions and identify potentially suspicious profiles. Trust Circle AI aims to bridge the gap between online networking and real-world interactions by providing personalized activity recommendations while incorporating mechanisms for identity verification and trust assessment. The platform supports activities including sports, study sessions, travel, hobbies, events, and social gatherings. Its architecture integrates user management, activity management, recommendation, trust evaluation, and safety monitoring into a unified platform.

**Keywords:** *Artificial Intelligence, Social Networking, Security, Recommendation Systems, Identity Verification, Smart Matching, Trust Score, Activity-Based Networking, User Safety, Machine Learning.*

## Honeypot That Learns from Attackers' Behaviour

[1]Shalini K V, [2]Nousheen P, [3]Manyashree M K
Dept. Artificial Intelligence and Machine Learning**,** T. John Institute of Technology, Bengaluru-83

**Abstract:** Cyber-attacks are getting harder to stop with detection alone, and defenders increasingly need systems that can act on what they see. A conventional honeypot lures attackers in and records what they do, which is useful for threat intelligence, but it behaves the same way no matter what the attacker does. Once an intruder suspects a trap, a static honeypot has no way to react. We propose an AI-based adaptive honeypot that closes this gap. The system will log attacker activity inside an isolated environment (scanning, login attempts, commands, file searches, credential hunting and privilege escalation), arrange it into time-ordered sessions, and use machine learning to estimate how far the attack has progressed and how risky it is. That estimate is not the end product. It feeds a response policy that decides whether to keep watching, plant decoy resources, restrict the session, isolate the container or end the session, always inside the honeypot boundary. A dashboard shows attack progress, risk and the system's reactions. The work is at an early prototype stage: this paper covers the problem, threat model, design, a first demonstration of the prototype on simulated traffic, its limitations and the evaluation plan, and measured results will follow once testing is complete. Our main question is whether behavior-aware adaptive deception gives better results than a static honeypot.

**Keywords:** *Adaptive Honeypot, Cybersecurity, Deception Technology, Intrusion Monitoring, Machine Learning, Threat Detection*

## CNN Based Smart Waste Segregation and Collection System

[1]Bhavana, [2]Nagveni, [3]Preeti

Dept. Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** In urban areas all over the world, waste management is becoming more and more important due to rising population levels and environmental sustainability objectives. In order to meet this challenge, the present paper introduces a waste separation and collection system based on a Convolutional Neural Network (CNN) which makes use of the capabilities of computer vision and machine learning. The system proposed is designed to automate the process of waste separation and to optimize the sorting of waste into bio-degradable and reusable categories. The system enables the real-time sorting of recyclable waste and trash by using a camera to take images of the waste as it reaches the various collection points and then moving waste to its appropriate collection point. The images are processed with a deep learning YOLOv8 model, this model identifying and classifying the waste items into bio-degradable, recyclable, and non-recyclable categories with a high degree a accuracy. This innovative waste management solution has the potential to transform waste management solution has the potential to transform waste collection and recycling procedures, making them more cost-effective, environmentally friendly, and suitable for sustainable waste management practices.

**Keywords:** *Waste management, Convolutional Neural Network, Recyclable, Biodegradable, Waste segregation, YOLOv8*

## AI-Enabled Detection of Exoplanets from Noisy Astronomical Light Curves

[1]Anirudh Sarkar, [2]Jaya Singhal, [3]Krishkumar Kalya, [4]Shashank Sharma, [5]Sharon Christa

[1-4] MIT Art, Design & Technology University, [5] MIT School of Computing

**Abstract:** Detecting exoplanets from astronomical observations is challenging because transit signals can be weak, noisy, incomplete, and affected by stellar variability or eclipsing binaries. Manual inspection of large volumes of light-curve data can be time-consuming and may overlook promising candidates. This paper proposes an artificial intelligence-enabled pipeline for automated exoplanet candidate screening using observations from NASA's Transiting Exoplanet Survey Satellite (TESS). The proposed system will combine data validation, preprocessing, noise reduction, periodic transit detection, feature extraction, and supervised machine learning classification. The Box Least Squares (BLS) algorithm will be used to identify periodic brightness reductions and estimate transit-related parameters. Random Forest and XGBoost will be investigated for distinguishing potential planetary signals from false positives. Evaluation will use precision, recall, F1-score, ROC-AUC, average precision, and confusion matrices. Dataset partitioning by target identifier will be used where applicable to reduce information leakage between training and test data. The planned system will also provide interactive light-curve visualization, candidate summaries, and confidence-based scoring. Its implementation will follow a staged development plan covering data preparation, transit detection, classification, and dashboard development. Since implementation and experiments remain pending, no measured performance results are claimed. The effectiveness of the proposed framework will be established through subsequent experimental evaluation.

**Keywords:** *Pattern Recognition, Astronomical Data Analysis, Machine Learning, Classification, Scientific Computing, Exoplanet Detection, TESS, Light Curves, Box Least Squares, Random Forest, XGBoost, Transit Photometry, False-Positive Classification.*

## Advances in Probability Distributions and Statistical Inference: A Review of Classical Foundations, Computational Advancements, and Applications in Machine Learning

[1]J. Raji, [2]Harsh Dwivedi, [3]Nicky Kumari, [4]Tejashwini Dalawai, [5]Manik Gir
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Statistics is in a place right now. The core probability axioms have stayed the same for fifty years. The way we work with them has shifted, mostly because computers can now do math faster. Since the hardware caught up the real changes have come not from rules but from new tools. Between 2023 and 2026 we'll see advances in how we infer distributions and make statistical inferences. These advances will be driven by computation not by theory. We're going to see more reliable and more consistent MCMC algorithms. We'll also see test cases that use thousands of shallow chains. We'll get regression methods that can handle dimensional noise. We'll see Gaussian processes applied to datasets without the data collapsing into just a few clusters. The literature on quantifying uncertainty in deep neural networks is growing fast. This review looks at four areas: Bayesian computation, high-dimensional and robust statistics, scalable Gaussian processes and uncertainty quantification in machine learning. It tries to pull what the current research actually says. There's no one method—only trade-offs. Most of these techniques are either more accurate and slower or more flexible and slightly biased. The best choice depends on the size and shape of your data. Classical statistics and machine learning are much closer than textbooks suggest. A neural network is a Gaussian process wearing a different hat. There are still open questions. What is real uncertainty, in a Bayesian net? How do we deal with stationary data? How can we do exact inference when sampling becomes too expensive? This review is meant to guide anyone moving from A-level probability into the AI systems they'll likely work with

**Keywords:** *Probability Distributions, Statistical Inference, Bayesian Inference, Markov Chain Monte Carlo, High-Dimensional Statistics, Non-Parametric Methods, Machine Learning, Uncertainty Quantification*

## Artificial Intelligence in Education: A Study of Opportunities, Challenges, and Responsible Use

[1]Shreyas S, [2]Manoj M, [3]Dileep Raju M, [4]Basavaraj N, [5]Arun V
Dept. Information Science and Engineering
T. John Institute of Technology, Bengaluru-83

**Abstract:** Artificial intelligence (AI) is increasingly being explored in education to support personalized learning, provide rapid feedback, assist teachers, and improve access to learning resources. This paper presents a structured review of the opportunities and challenges associated with AI-enabled educational tools. It considers adaptive learning systems, intelligent tutoring, automated assessment, and generative AI applications, while examining concerns related to privacy, bias, transparency, academic integrity, and unequal access. Rather than reporting an unperformed experiment, the paper proposes a practical framework for evaluating educational AI through learning support, usability, fairness, data protection, and teacher oversight. The discussion emphasizes that AI should complement educators and learner judgment rather than replace them. The paper concludes that responsible adoption requires clear institutional policies, appropriate evaluation, and meaningful human supervision.

**Keywords:** *Artificial intelligence, education, personalized learning, generative AI, educational technology, responsible AI*

## HealthID: Intelligent Patient Record Storage and Retrieval System

[1]Kanika Chalana, [2]Syed Faizan Mohammed Hussaini, [3]Mohammed Moinuddin, [4]Mohammed Farhan Javeed, [5]Sayed Bakar Sha Ali

Dept. of Artificial Intelligence and Machine Learning, T. John Institute of Technology

**Abstract:** In India, a patient's medical reports do not get saved in one place. Some may be in physical files, some may be laboratory PDF files and many are photographs of their medical reports saved in their mobile phones. So, if a new doctor asks for an earlier report, then the patient has nothing to show. If these reports are available in digital format, then the security for those reports would be inadequate, and these reports are difficult to retrieve by the person who cannot see the screen or the language of the report. In this paper, we propose a web application called HealthID which stores medical reports in the form of structured and encrypted reports and retrieves them via a unique HealthID or a QR code. The only authorized users who can upload the data are doctors and administrators; additionally, only medical reports in the PDF, JPG or PNG format can be uploaded after confirming the correct file type. Vitals, diagnoses, medications and laboratory values will be extracted using OCR and NLP, and all these reports will be encrypted using AES-256-GCM. Patients' login using password and OTP, and all the accesses to a report are restricted using role-based access. A voice assistant reads reports aloud for blind and visually impaired users, a language converter shows Hindi, Kannada, Telugu or Tamil beside the original text, and a help box sits on every page. HealthID is still a prototype, so this paper covers the design, the workflows and an evaluation plan instead of measured results.

**Keywords:** *Digital Health Records, HealthID, QR Code Access, AES-256-GCM Encryption, OTP Authentication, Role-Based Access Control, OCR, Multilingual NLP, Text-to-Speech, Accessibility*

## Autonomous Research Swarm: A Multi-Agent AI System for Automated Research

[1]Doddegowda B J, [2]Rachana P, [3]Sangamesh j [4] Sanjay, [5]Sharath A

Dept. of Information Science and Engineering

T. John Institute of Technology, Bengaluru-83

**Abstract:** An Autonomous Research Swarm is a decentralized, self-organizing artificial intelligence system in which multiple independent yet coordinated agents perform research tasks with minimal human intervention. Inspired by swarm intelligence, the system combines artificial intelligence, distributed computing, machine learning, and natural language processing to automate research activities. The proposed system consists of an Orchestrator that divides user queries into smaller tasks, a Browser Agent for data collection, a Coder Agent for calculations and analysis, an Adversarial Critic for validation, and a Consensus Layer that combines verified results into a structured final report. By enabling parallel processing, knowledge sharing, and automated verification, the system aims to improve research speed, accuracy, scalability, and efficiency while reducing human workload. The proposed approach has potential applications in scientific research, healthcare, education, cybersecurity, business intelligence, and other domains. However, challenges such as computational requirements, communication overhead, data security, and privacy must be addressed to ensure reliable operation.

**Keywords:** *Autonomous Research Swarm, Multi-Agent Systems, Swarm Intelligence, Artificial Intelligence, Distributed Computing, Natural Language Processing, Automated Research, Adversarial Validation, Consensus Mechanism, Parallel Processing*

## Applications of Graph Theory for Analysis of Social Networks

[1]J. Raji, [2]Sham HS, [3]Patil Bhagvan, [4]Siripireddy Umesh, [5]Chethan
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The paper "Applications of Graph Theory for Analysis of Social Networks" explains how graph theory can be used to study the relationships between groups, and organizations. In a social network, individuals are represented as nodes or vertices, while the connections between them are represented as edges. This representation helps us understand how people are connected and how information moves through a network. Nowadays, social networking such as Facebook, Twitter, and LinkedIn have millions of users who communicate and share information with one another. Graph theory provides a useful way to analyse these large networks and understand the relationships among their users. One of the important concepts in graph theory is centrality, which helps identify influential individuals and users who have many connections within a network. Different centrality measures can be used to understand importance of individuals based on their connections and positions in the network. Another important concept is community detection, which helps identify groups of users who are closely connected or share common interests. Graph theory can also be used to study information flow and understand how news, ideas, and messages spread from one individual to another. In addition, it helps researchers examine the structure of social networks and identify important connections between different groups. Dynamic graphs are useful for analysing to change over time as people form new relationships remove existing connections. Graph Networks can further support the analysis of complex networks by learning patterns from the relationships between connected users. These methods have applications in areas such as recommendation systems, the identification of influential users.

**Keywords:** *Graph Theory, Social Network Analysis, Centrality Measures, Community Detection, Information Diffusion, Dynamic Graphs, Graph Neural Networks, Recommendation Systems.*

## Development of a Portable Food Contamination Detection System for Rapid Food Safety Assessment

[1]Ramsagar Bhandari, [2]Vaishnavi M, [3]Nawaz, [4]Syed Yasir Arafat
Dept. Information Science and Engineering
T. John Institute of Technology, Bengaluru-83

**Abstract:** Food contamination is a major public health concern due to the presence of harmful microorganisms and chemical contaminants that may not be detectable through visual inspection. This research proposes the development of a portable, cost-effective food contamination detection system for rapid preliminary assessment of food samples. The proposed system will use suitable sensing technologies or rapid testing methods to identify specific contamination indicators and display the results in an understandable format. The primary objective is to provide a convenient solution for preliminary food safety screening outside laboratory environments. The system's performance will be evaluated in terms of detection accuracy, response time, portability, and reliability across selected food samples. This research aims to contribute towards accessible food safety assessment and support informed consumer decisions.

**Keywords:** *Food Contamination, Food Safety, Portable Detection System, Rapid Detection, Contaminant Analysis, Food Quality Assessment.*

## Deepshield: An AI-Powered Threat Detection Framework for Real-Time Cybersecurity Defense

[1] Pallavi T, [2]Amith Somanath Mali, [3]K S Manjunath, [4]S Varun, [5]Ravi Kiran
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** The escalating scale, speed, and sophistication of modern cyberattacks — including polymorphic malware, zero-day exploits, and multi-stage advanced persistent threats (APTs) — have exposed the limitations of traditional signature- and rule-based intrusion detection systems (IDS). This paper presents DeepShield, an AI-powered threat detection framework that combines deep sequence learning, unsupervised anomaly detection, and threat-intelligence correlation to identify malicious activity across network and endpoint telemetry in near real time. DeepShield employs a hybrid detection engine that fuses a convolutional-recurrent (CNN-LSTM) classifier with an Isolation Forest-based anomaly scorer, enabling it to detect both known attack signatures and previously unseen behavioral deviations. The framework further integrates a threat-intelligence enrichment layer aligned with the MITRE ATT&CK matrix and an automated response and orchestration module for adaptive mitigation. We describe the system architecture, data pipeline, and model design, and present an illustrative evaluation methodology using benchmark intrusion-detection datasets. In our simulated evaluation setting, DeepShield's ensemble approach outperformed standalone machine learning and signature-based baselines across accuracy, precision, recall, and false-positive rate. We conclude by discussing deployment considerations, limitations, and directions for future work, including federated training across distributed sensors and adversarial robustness testing.

**Keywords:** *Anomaly Detection, Cybersecurity, Deep Learning, Intrusion Detection System, Isolation Forest, MITRE ATT&CK, Threat Intelligence, CNN-LSTM*

## Mathematics In neural networks: Foundations of Learning, Optimization and Prediction

[1]J. Raji, [2] Hansika N [3]Saakshi M [4]Manasa N [5]V Vinesha
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Neural networks are widely used in artificial intelligence, but their learning process is fundamentally governed by mathematics. Linear algebra represents data and model parameters, calculus measures how changes in parameters affect predictions, probability supports uncertainty modelling, and optimization adjusts the network to reduce error. This paper presents an undergraduate-oriented mathematical overview of neural networks, focusing on the operations involved in a basic feedforward network and the role of backpropagation and gradient descent in training. A general methodology is proposed for studying these concepts through a small supervised classification experiment. The paper also discusses learning-rate selection, overfitting, activation functions, and evaluation measures. Rather than claiming experimental results, it outlines a reproducible procedure through which students can examine how mathematical choices influence model performance. The study aims to connect classroom mathematics with the practical design and training of neural-network systems.

**Keywords:** *Artificial Intelligence, Neural Networks, Mathematics, Machine Learning, Optimization, Classification, Linear Algebra, Calculus, Matrix Operations, Activation Function, Loss Function, Backpropagation, Gradient Descent, Neural Network.*

## Cardiovascular Disease Detection: A Hybrid Machine Learning-AI Framework for Personalized Diagnosis and Risk Assessment

[1]Kanika Chalana, [2]Vikas N, [3]Vamshi, [4]Sudarshan T N, [5]Shreyas H S
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Cardiovascular disease (CVD) remains a major challenge to global health, and accurate, personalized and easily understood diagnostic methods have therefore become more important than ever. This paper presents a hybrid machine learning and artificial intelligence framework that uses a Support Vector Machine (SVM) as the classifier, applies Particle Swarm Optimization (PSO) to tune the SVM hyperparameters, and employs SHapley Additive exPlanations (SHAP) to explain each individual prediction. The framework is built on four patient data vectors: medical imaging, clinical measurements, genomic information, and medical history, and the estimated risk can be updated whenever new patient information becomes available. The pipeline performs missing-value handling, feature scaling, categorical encoding and dynamic class weighting, followed by PSO-based optimization of the SVM regularization and kernel parameters and SHAP-based post-hoc explanation. The main experiments use MIMIC-III v1.4 with patient-level separation of the training, validation and test sets. The framework achieves 98.4% accuracy, 97.5% precision, 96.4% recall, 96.9% F1-score, 96.4% sensitivity, 98.7% specificity and a negative likelihood ratio of 0.036. The AUC-ROC is reported as 97.35% in the abstract and in the ablation table, and as 97.65% in Table IV; this difference is retained as originally reported. In addition to the results, this paper documents the methodology in detail, describes how each algorithm is used in the pipeline by comparing existing and proposed techniques, and includes a plagiarism and originality verification section.

**Keywords:** *Cardiovascular Disease, Machine Learning, Particle Swarm Optimization, Personalized Diagnosis, SHapley Additive exPlanations, Support Vector Machine*

## Explainable Student Churn Prediction Using Machine Learning and SHAP for Early Intervention

[1]Nishutha M S, [2]Deeksha T S, [3]Kavya, [4]Pavithra V, [5]Swathi G
Dept. of Information Science and Engineering, T. John Institute of Technology, Bangalore, India

**Abstract:** Student churn, the discontinuation of a programme before completion, is a persistent concern for educational institutions because it affects both academic outcomes and institutional performance. Most existing practice either relies on manual monitoring of marks and attendance, which identifies at-risk students too late, or on black-box machine-learning models whose predictions educators cannot question. This paper presents the design of an Explainable Student Churn Prediction System that converts academic and behavioural data, such as GPA, attendance and engagement, into a set of features, compares four machine-learning models (Logistic Regression, Decision Tree, Random Forest and XGBoost) and uses the best-performing model for prediction. SHAP (Shapley Additive Explanations) is applied to the fitted model so that every prediction is accompanied by a feature-level explanation of the factors behind a student's risk, and the results are presented through a dashboard with risk scores, explanations and recommendations for early intervention. The models will be evaluated using accuracy, precision, recall, F1-score and balanced accuracy; related studies report an F1-score of 0.840 and an accuracy of 98.9% on their own datasets. The system is intended as a decision-support indicator for educators, not as an automatic judgement about any student.

**Keywords:** *Student Churn, Dropout Prediction, Explainable AI (XAI), Learning Analytics, XGBoost, SHAP, Early-Warning System*

**Statistical Detection of data in Machine Learning Datasets**

[1]J. Raji, [2]B. Priya Darshini, [3]Deekshitha A, [4]Rachana R, [5]Vennela B
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Machine learning models learn patterns from the data provided to them, and thus the quality and fairness of the dataset can have a considerable impact on their performance. The bias in a dataset can be manifested in many ways, for example, if some groups of data are under-represented or if the distribution differs significantly between data groups. This paper proposes a statistical procedure for identifying potential biases in machine learning datasets. Hypothesis testing is used to analyze the distribution of selected features and evaluate differences between groups. The potential bias in the dataset may then be mitigated by applying appropriate data balancing procedures, after which the performance of the machine learning model can be evaluated. The work aims to demonstrate how statistical inference can be used to identify potential bias in data sets at an early stage and thus contribute to more accurate and reliable machine learning models. Selected concepts from the fields of probability, statistics, and machine learning were combined to form the basis of this work. Machine learning systems are increasingly being used for decision-making in areas such as healthcare, education, recruitment, finance, and public services. The reliability of these systems depends strongly on the quality and distribution of the data used for training. When a dataset contains unequal representation or systematic differences between groups, a machine learning model may learn about these patterns and produce different outcomes for different groups. This research presents a statistical approach for detecting potential bias in machine learning datasets. The study begins by identifying relevant groups and features within a dataset, followed by the calculation of statistical measures to examine their distribution. Hypothesis testing is then applied to determine whether the observed differences are statistically significant. When a significant imbalance is identified, an appropriate data-balancing technique is applied. A machine learning model is subsequently trained using both the original and improved datasets. Finally, model performance is compared before and after the bias-reduction process. The proposed approach provides a systematic way to move beyond simply stating that a dataset is biased. It uses statistical evidence to identify potential imbalances and evaluates whether reducing that imbalance affects machine learning performance. The study aims to support more reliable, transparent, and responsible development of machine learning systems.

**Keywords:** *Machine Learning, Dataset Bias, Statistical Hypothesis Testing, Statistical Inference, Data Balancing, Algorithmic Fairness, Bias Mitigation, Model Performance Evaluation.*

## Role of Statistical Methods in Healthcare Research

[1]J. Raji, [2]Angelina A, [3]Karthick CJ, [4]M Keerthana, [5]Jayanth, [6]Manoj B
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Statistical methods play a vital role in healthcare research by providing systematic techniques for collecting, organizing, analyzing, and interpreting medical and health-related data. These methods help researchers evaluate treatment effectiveness, identify disease patterns, understand risk factors, and assess the quality of healthcare services. Techniques such as descriptive statistics, sampling, correlation, regression analysis, hypothesis testing, and probability analysis support the accurate interpretation of research findings. Statistical methods also help compare treatment outcomes, evaluate clinical trials, predict disease trends, and support evidence-based medical decisions. The appropriate application of statistical techniques improves the reliability, validity, and accuracy of research results while reducing uncertainty and potential errors. Furthermore, statistical analysis assists healthcare professionals and policymakers in developing effective treatment strategies, allocating resources efficiently, and improving patient outcomes. This paper explores the importance of statistical methods in healthcare research, highlights their major applications, and emphasizes the need for accurate data collection, suitable statistical techniques, and proper interpretation of results. Overall, statistical methods are essential tools for advancing medical knowledge, improving healthcare delivery, and supporting informed decisions in clinical practice and public health.

**Keywords:** *Statistics, Healthcare, Research, Data, Analysis, Patients, Diseases, Treatment, Health, Medical, Results, Prevention.*

## STORELANE: A Phygital Commerce Platform for Real-Time Local Retail Discovery

[1] Siddu Tushara M S, [2]Anant Sharma, [3]George Lakra, [4]Harshit Singh, [5]Vipul Suthar.
Dept. of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Most small neighborhood retailers in India still trade offline. Shoppers who want a specific item often travel to a shop without knowing whether it is in stock, and the few shops that do appear online struggle to keep their listings correct. This paper presents StoreLane, a hyperlocal "phygital" platform that connects nearby customers with local vendors through product search, without asking shopkeepers to maintain numeric inventory. Each vendor listing has a three-state availability signal (green, orange, red) that weakens automatically on a 7, 30 and 50-day freshness schedule, so an unattended entry loses visibility instead of misleading the buyer. A customer can confirm an item by calling the vendor or opening a WhatsApp enquiry before travelling, and a temporary price-change notice covers the days after a vendor edits a price. The design is grounded in a review of fifteen recent studies on seller-side adoption barriers, information quality, visual trust cues, secure authentication and real-time web architecture.

**Keywords:** *StoreLane, Hyperlocal E-Commerce, Phygital Retail, Local Vendor Connectivity, Product Availability Tracking, Inventory-Free Retail Management, Information Freshness, Digital Transformation of Small Retailers, WhatsApp-Based Customer Enquiry, Trustworthy Product Listings*

## BEACON-Real-Time Crisis and Relief Orchestration Platform

[1]Shilpa N S, [2]Neha Fathima, [3]Sneha.M, [4]P.V. Pavitra, [5]Snehal Sahu
Department of Information Science and Engineering, T. John Institute of Technology, Bangalore

**Abstract:** Disasters such as floods, earthquakes, landslides, and industrial accidents continue to cause widespread damage to human lives and infrastructure, demanding faster and more intelligent emergency response mechanisms. Existing disaster management systems are largely fragmented, reactive, and dependent on manual coordination, leading to critical delays in communication and resource allocation. To address these limitations, this project proposes BEACON (Real-Time Crisis and Relief Orchestration Platform), an advanced disaster management system designed to enable real-time incident reporting, intelligent decision-making, and seamless coordination among victims, volunteers, and authorities. The platform integrates modern technologies including React JS, Node.js, Firebase, WebSocket, MQTT, YOLOv8, Llama-3, GPT-4, and Twilio to deliver low-latency alerts, AI-based verification, priority-based emergency handling, and multi-channel communication support. Dedicated dashboards for each stakeholder ensure role-based access and real-time situational awareness, while geolocation and mapping features enable precise tracking of disaster-affected areas. The system is designed to function reliably even in low-connectivity environments through SMS-based fallback communication. By combining artificial intelligence, IoT integration, and cloud infrastructure into a unified platform, BEACON transforms disaster response from a reactive and fragmented process into a proactive, coordinated, and data-driven ecosystem. The proposed system aims to significantly reduce response time, improve resource utilization, and ultimately contribute to saving lives and building resilient communities.

**Keywords:** *Disaster management, Real-time, Communication, Authorities, Emergency.*

## CareLoop: Caregiver-Linked Medication Reminder App

[1] Pallavi T, [2]Sobia Fatima, [3]Rabiya Basree, [4]Keerthana A S
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Medication adherence is a major problem faced by many users including children, independent adults and the elderly. Current reminder apps often have a one-size fits all approach, require constant internet access and offer poor protocols for when a user misses a dose.

This paper presents CareLoop, a context aware medication adherence app which caters to different patient types. The app can be used in child, adult and elderly modes. Each mode adapts how the user interface, alert system and escalation protocols for missed doses are presented. The app has been designed to work offline. This means that the alarm and dose tracking systems are stored at device level. The cloud is only used to allow multiple users, link caregivers and escalate alerts. The app also features a refill system based on counting the number of pills. It will automatically alert caregivers but won't change the dose. This provides a reliable and secure solution to improve adherence.

**Keywords:** *Caregiver Alerts, Firebase, Local-First Scheduling, Medication Adherence, Medication Reminders, Missed-Dose Escalation, Pill-Count Tracking, Privacy, Safety, User-Centered Design*

## AI Powered Technical Support Tool (FixiQ)

[1]Shilpa N S, [2]Dakshatha N, [3]Deekshitha H S, [4]Manasa N
Dept. Of Information Science and Engineering, T. John Institute of Technology, Bangalore

**Abstract:** Technical support systems frequently depend on manual troubleshooting, ticket-based workflows, and static frequently asked questions, which can result in delayed responses, repetitive workload, and inconsistent solutions. This paper presents FixiQ, an AI-powered technical support tool designed to provide fast, intelligent, and automated assistance for common technical problems. The proposed system accepts natural-language queries through a user-facing chatbot, processes the input using Natural Language Processing (NLP), classifies the issue using supervised machine learning, and retrieves a suitable solution from a structured knowledge base. The system identifies technical issues across network, software, and hardware categories using a Naive Bayes classification approach, followed by search and matching for solution selection. FixiQ also incorporates authentication, administration, notification, and feedback modules, while complex or unresolved problems can be escalated to human support. The implementation uses Python for AI and machine-learning processing, Node.js for backend services, React.js for the frontend, MongoDB/JSON for data storage, and OpenAPI integration for external services. Reported project evaluation indicates response times below one minute and automated handling of approximately 70–80% of customer interactions without human assistance. These figures are reported project outcomes and require controlled evaluation for independent validation.

**Keywords:** *FixiQ, AI-Powered Technical Support, Natural Language Processing (NLP), Machine Learning, Naive Bayes Classifier, Automated Troubleshooting, Intelligent Chatbot, Knowledge-Based Solution Retrieval, Technical Issue Classification, Human Support Escalation.*

## Food Calorie Detection and Estimation Using Machine Learning

[1]Pallavi T, [2]Sakshi Jayanth, [3]Manisha Srinivas
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Deep Learning for Automated Food Calorie Estimation from Images is a key Computed Dietary Assessment method. Standard nutritional assessment relies very largely on manual food recording, portion size estimation and nutrient look-up, all of which are extremely time-consuming and susceptible to large estimation errors. To surmount these constraints, we propose this paper as a unified deep learning framework capable of automatically localizing multiple objects on-site via YOLOv8-based models, handling fine-grained classification based on CNN/ResNet and automating mapping into nutritional databases. The architecture in question also uses YOLOv8 to identify simple and overlapping food items in complex tablescapes, as well as region cropping to remove distracting background. Then, a ResNet classifier similar to the neural architecture of image classification is used to recognize food categories and maps predicted category labels back to nutritional data tables for total calories calculation. Comparative literature analysis isolates this hybrid integration of single-stage bounding-box detection as a competitive standalone deep residual classifier; we report superior localization, precision and reduced estimation error relative to the state-of-the-art across a range of performance metrics.

**Keywords:** *Automated Food Recognition, Calorie Estimation, Convolutional Neural Networks, Deep Learning, Food Image Analysis, Nutritional Assessment, ResNet, YOLOv8*

## Probability-Based Spam Detection Using Statistical Inference and Machine Learning

[1]J. Raji, [2]Shreekeerthi N., [3]Tanuja Y. L., [4]Spoorthi M P, [5]Manasa M
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The explosive growth of the modern communications sector has created a significant increase in spam and junk messages, making it challenging to separate garbage and non-garbage data. Contemporary approaches to filtering such information may have shortcomings, including an inability to adapt to changing conditions. This paper scopes out the development and implementation of a probability-based message detection system that employs statistical inference and machine learning algorithms to detect spam information. The proposed solution involves analyzing message content and using statistical inference to determine the likelihood of spam. Additionally, the process employs data mining methods to extract features from text, including tokenization and stop-word removal, and machine learning to classify texts with high accuracy. The implementation results demonstrated that it could classify spam and non-spam messages with increased accuracy. The approach reduces false-positive classifications and enhances the overall learning ability of the system. Therefore, this study's findings provide a solid foundation for developing intelligent text detection and mining solutions. This research provides a significant insight into developing an efficient spam detection system that utilizes the power of statistical inference and machine learning to separate spam and non-spam information. The findings demonstrate that such an approach could be applied to build reliable computer systems that could automatically filter messages without the need for extensive human input.

**Keywords:** *Spam, Ham, Filtering, Probability, Machine Learning, Statistics, Data Mining, Tokenization, Stopword Removal, Accuracy, False Positives, Automatic System, Spam Detection, Statistical Inference, Probability-Based Classification, Naïve Bayes, Text Mining, Email Filtering, Feature Extraction, Data Classification, Cybersecurity.*

## MediScan: An AI-Powered Personalized Medication Intelligence and Safety Platform

[1]Asha S Manek, [2]Anushree Kowru, [3]Swathy Kowru
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Medication errors often occur due to incorrect identification and accidental duplication. This paper introduces MediScan, a web based clinical support tool which allows prescriptions and physical objects to be converted into individualised, diet aware medication plans. Computer Vision and OCR is used to extract information from images of tablets, capsules, syrups, labels and QR code's. Confidence scores are calculated by combining information from different sources to identify uncertain detections. Drug information is collected from a validated database ensuring there is a clear distinction between accurate drug information and the advice given to users. The tool includes functionality for medication reminders based on meals, duplication alerts based on rules identifying medications containing the same active ingredient but different brand names, and clear disclaimers for warnings with sources. The tool works through a doctor centred process providing a reliable solution to improve medication safety and compliance in patients.

**Keywords:** *Confidence Estimation, Duplication Detection, Healthcare Application, Meal-Aware Scheduling, Medication Safety, OCR, Pill Recognition, Prescription Digitization.*

## A Digital Twin-Driven Multi-Agent Rag Framework for Personalized Learning Through Knowledge Gap Detection and Adaptive Content Recommendation

[1]Meghana R, [2]Muskan B, [3]Nithya Shree J
Dept of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Generative Artificial Intelligence (GenAI) has created new opportunities for developing intelligent and personalized learning environments. Existing studies on Generative AI-driven multimodal teaching, including approaches based on BERT optimization, primarily focus on improving content delivery and teaching effectiveness within specific educational domains. However, limited attention has been given to continuously modeling individual learners, detecting their specific knowledge gaps, and adapting learning resources according to their changing needs. This paper proposes a Digital Twin-Driven Multi-Agent Retrieval-Augmented Generation (RAG) Framework for Personalized Learning Through Knowledge Gap Detection and Adaptive Content Recommendation. The framework maintains a dynamic Digital Twin of each learner using learning performance, assessment results, interaction history, strengths, weaknesses, and learning preferences. Multiple specialized agents collaborate to analyze learner information, identify knowledge gaps, retrieve relevant educational resources, generate personalized content, and recommend suitable learning activities. The RAG component grounds generated responses in a curated educational knowledge base to improve the relevance of learning content. The learner model is continuously updated based on subsequent assessments and interactions, enabling adaptive learning rather than static content delivery. The proposed framework aims to provide individualized learning support by integrating learner modeling, knowledge-gap detection, retrieval-based generation, and adaptive recommendation within a unified architecture.

**Keywords**: *Digital Twin, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Generative Artificial Intelligence, Personalized Learning, Knowledge Gap Detection, Adaptive Content Recommendation.*

## AI-Based Large Language Models for Analyzing Code Security and Identifying Data Leakage Risks

[1]Shalini K V, [2]Ryan Jacob Charanghat, [3]Varun Suresh Babu, [4]Rithwik Pramod, [5]Chendu K
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Large Language Models (LLMs) are increasingly being incorporated into software engineering practices with a significant increase in productivity. However, they create new risks in cybersecurity, such as insecure code generation, prompt injection, information leakage, and erroneous mitigation suggestions. Static analysis methods struggle to identify and detect dynamic context vulnerabilities and also have increased false positives. Retrieval augmented pipelines are also susceptible to inference-based information leakage.

This paper presents SecureLLM-CodeShield. It is a security framework that detects vulnerabilities in source code when using LLMs. The framework includes real-time source code analysis, credential discovery, automatic privacy protection, and prompt injection protection. Through experimentation against traditional vulnerability detection methods we show SecureLLM-CodeShield increases detection by 6.8%, reduces false positives by up to 31%, and reduces information leakage of sensitive data by 52%. This provides a strong defence for secure AI engineering.

**Keywords***: Data Leakage, Large Language Models, Prompt Injection, Retrieval Security, Secure Code Analysis, Software Security, Vulnerability Detection*

## Gamified Territory Based Engagement Platform for Active and Healthy Communities

[1]Kanika Chalana, [2]Jupalli Sinduja, [3]K. Ashritha, [4]Lasya. K, [5]Keerthana. S
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract**: The rapid growth of mobile health (mHealth) technologies and wearable devices has transformed the way individuals monitor their physical activity; however, many existing fitness tracking systems suffer from limitations such as inaccurate activity recognition, high energy consumption, and lack of personalized feedback. This paper proposes a GPS-based fitness tracking system that integrates smartphone sensors, wearable devices, and machine learning algorithms to monitor physical activities in real time. The system utilizes GPS for location tracking and motion sensors such as accelerometers and gyroscopes for activity recognition, while a cloud-based architecture is used for data storage and analysis, enabling real-time feedback and performance tracking. The proposed system improves activity recognition accuracy and enhances user engagement by providing personalized insights, demonstrating the effectiveness of combining GPS, wearable technology, and machine learning in developing intelligent fitness monitoring systems. Furthermore, the system emphasizes scalability and adaptability for real-world applications, making it suitable for healthcare monitoring, sports training, and daily fitness management. Overall, this work highlights the potential of advanced mobile and wearable technologies in promoting preventive healthcare and supporting data-driven decision-making for improved user well-being.

**Keywords:** *Activity Recognition, Fitness Tracking, GPS, IoT, Machine Learning, mHealth, Wearable Sensors*

## SMART DEVELOPER ASSISTANT: AI-Based Automated Code Review and Quality Analysis System

[1]Siddu Tushara M S, [2]Ananya K M, [3]Mohan C, [4]Praveen P, [5]Rakesh Gowda S
Dept. Of Information Science and Engineering, T. John Institute of Technology Bangalore, India

**Abstract:** Code quality is an important part of software development, but manually reviewing every line can be time-consuming. This paper presents Smart Developer Assistant, a web-based prototype designed to support code review by combining a React.js interface, a Node.js and Express backend, and ESLint-based static analysis. The system accepts source code, runs configured linting rules, and presents diagnostics with rule identifiers and source locations. It also provides understandable explanations and improvement suggestions, and supports automatic fixes for rules for which ESLint provides a fix. The displayed quality summary is intended to help developers prioritise findings; it is not a measure of program correctness. The prototype is evaluated through functional test scenarios, including source submission, rule detection, diagnostic display, recommendations, and supported auto-fix behaviour. Quantitative accuracy or performance claims are not made because a controlled benchmark evaluation has not yet been completed. The system is intended to assist developers and does not replace testing, security review, or human judgement, automated code review, software quality, static analysis, ESLint, code smells, code diagnostics, developer tools, automated fixing.

**Keywords:** *Automated Code Review, Software Quality, Static Code Analysis, ESLint, Smart Developer Assistant, Code Diagnostics, Code Smell Detection, Automated Code Fixing, Developer Tools, Software Development*

## Fake and AI-Generated Detection System for Multimodal Data

[1]Madhavi Dixit, [2]Avinash H, [3]Chetan Kumar C, [4]Dhyan H, [5]Sujay Devadiga
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The rapid development of generative artificial intelligence technology has made it possible to create very realistic images that are almost indistinguishable from real-life visuals. The dissemination of AI-generated images through social media platforms has made it possible for disinformation through visual data and made it challenging to verify online information. The paper introduces a system for cross-lingual detection of AI-generated disinformation that concentrates on analyzing the images without consideration of the language used to describe the images. The system utilizes deep learning and transfer learning on a pretrained ResNet18 model to classify images into two categories: real and AI-generated. A dataset consisting of real images and AI-generated images is employed along with preprocessing steps such as resizing, normalization, and data augmentation. The data is split into train and test data, and the model is trained using the Adam Optimizer and Cross-Entropy Loss. The performance of the system is measured based on accuracy, precision, recall, F1 score, and confusion matrix. For better interpretability, the Grad-CAM algorithm is used to highlight the areas of the image which play an important role in the decision made by the model. The proposed system gives not only the output label of the detected class but also the confidence of the output. This work presents the possibility of using the deep learning on images for detecting AI-based visual disinformation.

**Keywords**: *AI-Generated Disinformation, Artificial Intelligence, Computer Vision, Cross-Lingual Detection, Data Augmentation, Deep Learning, Digital Forensics, Explainable AI, Fake Image Detection, Grad-CAM, Image-Based Analysis, Image Classification, Image Preprocessing, ResNet18, Social Media Analysis, Synthetic Images, Transfer Learning, Visual Misinformation*

## Mindful Me: A Multimodal AI-Assisted Framework for Mental Well-Being Support

[1]Kannan Senthamurai, [2]Jeeshma K, [3]Mopuram Janeethya, [4]Sara Farhath, [5]Harshitha R, [6]Swathi KN
Dept. of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Finding support for emotional well-being can be hard when a person does not know where to start, fears stigma, or cannot locate a suitable professional. This paper describes Mindful Me, a web platform that places self-reflection, screening information, guided self-care, and professional appointment booking in a single user journey. The application combines daily mood and stress check-ins, sleep and energy tracking, PHQ-9 and GAD-7 questionnaires, journaling, a conversational companion and rule-based recommendations. A separate safety component examines each incoming message before it reaches the conversational module and, when worrying language is found, points the user to configured support resources. Users, mental-health professionals, and administrators work through separate role-based interfaces, and a professional profile is shown as verified only after administrator approval. The planned implementation uses React, TypeScript, and Tailwind CSS for the interface and FastAPI, SQLAlchemy and PostgreSQL on the server. Screening scores are presented as indicators for self-reflection, not as diagnoses. This prototype description does not establish clinical effectiveness, and voice and facial-expression analysis remain possible future extensions rather than verified capabilities. The paper records the design, workflow, safety considerations, and evaluation plan of the platform.

**Keywords:** *Mental Health Support, Emotional Well-being, Mindful Me, Digital Mental Healthcare, Mood Tracking, PHQ-9 and GAD-7 Screening, AI Conversational Companion, Personalized Self-Care, Crisis Detection, Role-Based Access Control.*

## Brain Tumor Prediction Using Deep Learning

[1] Amrita Sinha, [2] Abhishek S, [3] Darshan M R, [4] Harshitha R, [5] Monisha Sindu C
Dept. of Information Science and Engineering,
T. John Institute of Technology, Bengaluru-83

**Abstract:** Brain tumors are among the serious neurological conditions that require timely and accurate detection to support appropriate clinical decision-making. Magnetic Resonance Imaging (MRI) is one of the most commonly used imaging techniques for examining brain structures because of its ability to provide detailed information about soft tissues and abnormal regions. However, manual examination of MRI scans can be time-consuming and depends heavily on the experience of medical professionals. Variations in tumor size, shape, intensity, and location further increase the difficulty of consistent image interpretation. To address these challenges, this work proposes a deep learning-based framework for automated brain tumor classification using Convolutional Neural Network (CNN), VGG16, and ResNet50 architectures. The proposed system consists of MRI image preprocessing, feature extraction, classification, model comparison, and explainable prediction. The preprocessing stage includes image resizing, normalization, and data augmentation to obtain consistent input and improve model generalization. A custom CNN is used as a baseline model, while VGG16 and ResNet50 are employed as transfer-learning architectures for extracting deeper image representations. The models classify MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor. Model performance is evaluated using accuracy, precision, recall, F1-score, and confusion-matrix analysis. The best-performing architecture is subsequently considered for final prediction, while Grad-CAM is incorporated to visualize image regions that influence the model's decision. The proposed framework demonstrates the potential of deep learning as a computer-aided approach for brain MRI classification and provides an interpretable prediction mechanism that can support, but not replace, professional medical analysis.

**Keywords:** *Brain Tumor Classification, Magnetic Resonance Imaging (MRI), Deep Learning, Convolutional Neural Networks (CNN), Transfer Learning, VGG16, ResNet50, Grad-CAM, Explainable Artificial Intelligence (XAI), Medical Image Analysis*

## Automated Compliance Verification of Packaged Commodities Under the Legal Metrology (Packaged Commodities) Rules, 2011 Using Image and Label Analysis

[1]Asha S Manek, [2]Trumbakeshwar Datt, [3]K Bhimesh, [4]Raghavendra V, [5]Aman Yadav
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Products which are sold through retailers, grocery stores and online portals in India need to have labels according to the provisions of the Legal Metrology Act, 2009 and Legal Metrology (Packaged Commodities) Rules, 2011. This includes the name and address of the manufacturer, packer or importer, the net quantity, the maximum retail price, the month and year of manufacture/packaging and consumer care information. Since the number of products is too many to manually inspect all products, only a sampling is done and thus problems with missing labels, fonts and price information can easily be overlooked. The software solution proposed in this paper can automatically extract text from the product label and catalog pictures and check it for compliance with the 2011 Rules. This includes text localization, optical character recognition, compliance rules engine, font size and readability validation and a dashboard powered by a repository.

**Keywords:** *Compliance Monitoring, E-Commerce Compliance, Font-Size Detection, Label Inspection, Legal Metrology, Optical Character Recognition, Packaged Commodities, Regulatory Compliance, Rule-Based Validation, Text Detection*

## Role of Statistical Methods in Healthcare

[1]G. Bhavya, [2]Lathika Reddy B R, [3]Dhruva G, [4]Lakshmishree R V, [5]J. Raji
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** This paper presents a comprehensive review of the role of statistical methods in healthcare research, with particular emphasis on their applications, limitations, and contribution to evidence-based medical decision-making. Statistical methods enable healthcare researchers to systematically collect, organize, analyse, and interpret medical data and are essential for identifying disease patterns, evaluating treatment effectiveness, estimating health risks, and drawing reliable conclusions from clinical and public health studies. Although previous studies have examined individual statistical techniques and their applications in specific healthcare domains, there remains a need for an integrated understanding of how different statistical methods collectively support various stages of healthcare research. This paper addresses this gap by comparatively examining descriptive statistics, inferential statistics, hypothesis testing, regression analysis, and survival analysis and by identifying the practical challenges associated with their application. Evidence from the reviewed studies indicates that different statistical methods serve complementary purposes. Descriptive statistics summarize patient and population characteristics, inferential methods support generalization from samples to populations, regression models identify relationships and risk factors, and survival analysis evaluates time-to-event outcomes. The review also identifies data quality, missing observations, inadequate sample sizes, inappropriate model selection, bias, and incorrect interpretation as important factors affecting the reliability of healthcare research. The paper further highlights the growing integration of traditional statistical methods with machine learning and data-driven healthcare technologies.

**Keywords:** *Healthcare Research, Statistical Methods, Data Analysis, Evidence-Based Medicine, Clinical Trials, Regression Analysis, Hypothesis Testing, Survival Analysis, Epidemiology*

## AURORA-FX: Intelligent Surveillance Forensics Using AI-Based Evidence Extraction, Correlation and Data Recovery

[1] K V Shalini, [2]Karthik S K, [3]Rohith P, [4]Jyothis Joshy, [5]Sindhu M Telgar
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Forensic evidence from DVR and NVR surveillance footage is becoming more commonly used in criminal and civil cases. However, different vendors have their proprietary formats of storing videos, file systems and compression techniques, which requires investigators to operate with multiple tools specific for each vendor separately. This increases investigation time, creates potential chain-of-custody issues and complicates standardization efforts among vendors. In this paper, we introduce AURORA-FX – an architecture, vendor-independent and modular framework, which unites DVR/NVR forensics in a single six-stage workflow: Acquisition, Recovery, Reconstruction, Analysis, Verification and Reporting. AURORA-FX is not introducing any new lower-level algorithms but rather offers standardization of evidence acquisition process, recovering of deleted and fragmented video, chronological reconstruction, modular video analysis, integrity verification, generation of evidence graphs and report production within one framework. The architecture, its components and evaluation plan for recovery effectiveness, reconstruction accuracy and performance of processing operations are described in this paper. Due to the early design and prototyping stage, no experiment results are provided in this paper.

**Keywords:** *Chronological Reconstruction, Cryptographic Integrity Verification, Digital Forensics, DVR/NVR Forensics, Evidence Acquisition, Evidence Recovery, Modular Framework, Video Analysis*

## Forecasting Cryptocurrency Price Prediction Using Long Short-Term Memory Networks

[1]Doddegowda B J, [2]Akshatha N H, [3]Pavani L, [4]Srushti Shettar, [5]Uday Kiran B R
Dept. of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Cryptocurrency markets experience extreme volatility and non-stationary price dynamics driven by speculative trading, macroeconomic developments, and investor sentiment, making accurate future price forecasting an exceptionally challenging task. Conventional statistical and machine learning models, such as ARIMA, Support Vector Regression, and Random Forest, struggle to model complex long-range temporal dependencies inherent in sequential financial series. In this work, we present an end-to-end Long Short-Term Memory (LSTM) recurrent neural network framework for sequential cryptocurrency price prediction. Daily historical transaction data containing Opening, High, Low, and Closing (OHLC) prices alongside trading volumes are collected, cleaned, normalized via Min-Max scaling, and transformed into overlapping sliding-window temporal sequences. The LSTM model captures nonlinear sequential trends through gated memory states. Experimental evaluation on unseen historical test data demonstrates that the proposed LSTM framework achieves a Mean Absolute Error (MAE) of 1.7421, a Mean Squared Error (MSE) of 5.2838, a Root Mean Squared Error (RMSE) of 2.2986, and a Mean Absolute Percentage Error (MAPE) of 3.0037%, consistently outperforming conventional baseline estimators.

**Keywords:** *Cryptocurrency Price Prediction, Long Short-Term Memory (LSTM), Deep Learning, Time Series Forecasting, Cryptocurrency Market Volatility, Sequential Data Analysis, OHLC Data, Min-Max Normalization, Financial Time Series, Model Performance Evaluation*

## MindgaurdAI

[1]Siddu Tushara M S, [2]Bindu Priya N, [3]Karthika Shree S, [4]Kaveri H S, [5]Sahana H
Dept. of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Psychological distress among undergraduates frequently remains invisible to institutions until it has become severe, because campus counselling is largely appointment-driven, stigma discourages help-seeking, and consumer wellness applications sit outside institutional care pathways. This paper presents MindGuard AI, a privacy-preserving web platform that couples a calm, student-facing wellness portal with a counsellor triage dashboard and an administrative governance view. Students complete a seven-step onboarding that records stress triggers, recovery tendency, grounding preferences and explicit consent, and thereafter use mood journaling, a conversational companion, regulation exercises, cognitive-reframing and bubble-popping mini-games, and a non-punitive wellness garden. Journal text is sanitised and sealed with AES-256-GCM, passwords are protected with crypt, and access is governed by JSON Web Token authentication with student, counsellor and administrator roles. A transparent Student Distress Index (SDI), combining mean mood valence, seven-day mood velocity and a crisis-keyword flag, drives a three-tier triage rule; counsellors receive only salted SHA-256 pseudonymous identifiers, scores and trends, never journal prose. The implementation uses React, Node.js with Express, PostgreSQL 16, Redis 7.2 with BullMQ, and a Python PDF service. Synthetic load tests with up to 1,000 virtual users sustained 2,400 requests per second at a mean latency of 114.6 ms, a 99th-percentile latency of 182.1 ms and a 0.05% error rate. The index is rule-based and awaits clinical validation through an ethically approved pilot.

**Keywords:** *Student Mental Health, Clinical Triage, Role-Based Access Control, Pseudonymisation, Distress Index, Digital Wellness.*

# Pairing Artificial Intelligence with Brain–Computer Interfaces: An Empirical Case Study of Deep Learning for Imagined-Speech Alphabet Decoding and the Evaluation Pitfalls of AI-Driven BCI

[1]Kannan Senthamurai, [2]Mohammed Junaid Khan, [3]Sanjana S, [4]Nirmitha Druthi
Dept. of Information Science and Engineering, T. John Institute of Technology, Bangalore

**Abstract:** The central idea of this work is the pairing of artificial intelligence and machine learning (AI/ML) with brain–computer interface (BCI) technology: using learned models to translate raw neural activity directly into symbols, commands, and words, without recourse to muscular output. This paper examines that pairing empirically through a complete AI-driven BCI for imagined-speech alphabet decoding. The system processes 690 EDF recordings from 23 subjects acquired on a 14-channel Emotive EPOC headset at 128 Hz, extracts a 322-dimensional hand-crafted feature vector per trial, and trains a 427,522-parameter residual one-dimensional convolutional network across three stimulus paradigms (Char, Digit, Image), deployed through an authenticated Streamlit application that converts uploaded recordings into spelled words. We reproduce every reported result exactly and verify that the deployment path is numerically faithful to training to float32 precision. We then interrogate whether the AI is learning what the BCI claims to decode. Four controlled experiments show that it is not: a Gaussian-noise augmentation written into the dataset before the train/test split leaves 99.8% of the test rows with a near-duplicate in training, and the reported 95.65% accuracy is a weighted average of 100% on twinned rows and 50% on the twelve rows without a twin. Under trial-grouped and subject-grouped protocols, six classical classifiers and the project's own CNN all fall to chance or below, while the identical features decode subject identity at 92–98% against a 4.35% chance level. The lesson generalizes beyond this project: when AI/ML is grafted onto BCI, the model will readily learn the recording session rather than the thought, and conventional random-split evaluation will certify it. We close with a set of design principles for AI-driven BCI covering augmentation placement, grouped validation, subject normalization, artefact handling, and deployment contracts.

**Keywords:** *Brain–Computer Interface (BCI), Imagined Speech Decoding, Neural Signal Processing, EEG Signal Classification, Data Leakage, Subject-Independent Validation, Model Generalization, Robust Evaluation, Neural Signal Artifacts.*

## A Context-Aware Hybrid Recommendation and Multi-Criteria Itinerary Generation Framework Using Distributed Microservices

[1]Amrita Sinha, [2]Akash D Patil, [3]Hemanth Kumar L, [4]Murgan S, [5]K Sai Teja .

**Abstract:** Recommending personalized social outings and coherent multi-stop itineraries requires synthesizing multidimensional user preferences, real-time contextual dynamics, geospatial travel constraints, and diverse group consensus. Conventional recommendation engines typically treat venue suggestions as isolated, single-objective prediction tasks, failing to account for temporal sequencing, occasion-specific semantics, ambient weather conditions, and review sentiment. In this paper, we propose a context-aware hybrid recommendation and dynamic itinerary planning architecture engineered for social outings and group activities. The framework is instantiated upon an event-driven, microservices-based distributed topology that decouples online real-time inference from asynchronous sentiment processing and continuous model training. The core algorithmic engine implements a multi-criteria scoring and rank-fusion mechanism that jointly models geospatial proximity via Haversine geofencing, occasion compatibility vectors, real-time weather penalty coefficients, and sentiment polarity extracted from user reviews. Furthermore, an automated timeline sequencing algorithm formulates itinerary generation as a time-windowed combinatorial optimization problem, integrating group preference aggregation via social choice consensus models to eliminate intra-group utility disparity. We empirically evaluate the proposed system using an offline benchmark evaluation framework and simulated multiagent interaction logs across urban point-of-interest catalogs. The experimental results demonstrate that our hybrid fusion pipeline achieves an NDCG@10 of 0.842 and a Mean Reciprocal Rank (MRR) of 0.789, outperforming baseline collaborative filtering and standalone content-based rankers by 18.4% and 12.6%, respectively. System latency benchmarks show an end- to-end P95 response time of 74.2 milliseconds across Dockerized microservices under sustained concurrent loads. The proposed architecture bridges the gap between context-aware machine learning and production-grade distributed systems, delivering scalable, contextually cohesive, and group-optimized itinerary generation for real-world outing planning.

**Keywords:** *Context-Aware Recommendations, Itinerary Planning, Multi-Criteria Decision Making, Microservices Architecture, Sentiment Analysis, Group Consensus*

## AI Resume Parser and Job Recommendation System

[1]Siddu Tushara, [2]Bharath Gowda M, [3]Chandana C K, [4]Deepak Kumar Thakur L, [5]Lisha Ramesh
Dept. of Information Science and Engineering, T. John Institute of Technology, Bengaluru-83

**Abstract:** Finding suitable employment in a competitive job market is difficult, as manual searching and resume screening are slow and inefficient. This project proposes an AI-based system that extracts skills, education, qualifications and work experience from resumes using NLP (spaCy, ResumeParser) and matches them with job descriptions using TF-IDF vectorization, cosine similarity and Nearest Neighbors. Job data collected from online sources is stored in structured datasets. The system is a Flask web application where users upload a resume and view personalized job recommendations with links to the postings. It does not guarantee employment or perfectly accurate parsing, but it narrows the search and makes matching faster and more consistent, contributing to SDG 8 through better access to productive employment and to SDG 9 through an innovative digital recruitment system.

**Keywords:** *Resume, parser, NLP, Job data, Education, resume*

## AI-Based Plant Disease Detection

[1]Netravathi S, [2]Nitturu Ishwarya, [3]Abhishek M G, [4]Jayachandra Reddy, [5]Naveen Kumar
Dept. Artificial Intelligence and Machine Learning
T. John Institute of Technology, Bengaluru-83

**Abstract:** Plant disease outbreaks present a serious threat to the world's agricultural production, economy, and food security. Manual inspections are highly laborious, error-prone, and inefficient, hence computer vision solutions such as deep neural networks, particularly Convolutional Neural Networks (CNN), are gaining great interest in the classification task. However, the accuracy of such algorithms is extremely inconsistent between a controlled academic environment and a large variety of real-world conditions. This paper proposes an automated solution to detect plant diseases in the field. The solution implements a fine-tuned CNN (ResNet50 / MobileNetV2) and transfer learning to extract the relevant features of a leaf, such as shape, color, and texture. The system is trained on a mixed set of public (PlantVillage) and private leaf images with extensive augmentation to simulate various backgrounds, light conditions, and camera properties in order to generalize well on different inputs. The proposed solution is a mobile-friendly, low-cost application that does not require cloud computation. Experiments performed to reach over 90% classification accuracy with the presented architecture suggest that accounting for a variety of devices and noise during the training phase significantly increases the accuracy, therefore enabling better resource allocation with the help of an automatic plant disease detection system.

**Keywords:** *Artificial Intelligence, Deep Learning, Convolutional Neural Networks (CNN), Plant Disease Detection, Transfer Learning, Mobile Agriculture, Precision Farming, Image Classification.*

## Secure Enhancement for Forensic Evidence using Blockchain Technology

[1]Shilpa N.S, [2]Janani .J, [3]M.Manish Kumar, [4]Mahesh D.S, [5]Sheshadri G.M
Dept. of Information Science and Engineering
T. John Institute of Technology, Bengaluru-83

**Abstract:** Forensic evidence is useful in court only if its authenticity and integrity can be demonstrated and every handover from collection to presentation is documented. Conventional chain-of-custody records are kept on paper or in centralised databases, where entries can be altered, lost or disputed. This paper proposes a blockchain-based system for the secure management of forensic evidence. Each evidence file is hashed with SHA-256 and encrypted with AES-256, and the ciphertext is kept in off-chain storage. The hash, the storage reference, case metadata and every custody transfer are recorded on a permissioned blockchain through smart contracts, while role-based access control restricts who may register, transfer, access and verify evidence. A verification module recomputes the hash of retrieved evidence and compares it with the value stored on the blockchain, so that any modification is detected. The evaluation plan covers tamper-detection rate, transaction latency, gas cost, throughput and storage overhead. No experimental results are claimed in this paper.

**Keywords:** *Forensic, Blockchain, chain-of-custody*

## Smart Helmet

[1]Harsh Dwivedi R, [2]Tejashwini Dalawai, [3]Nicky Kumari, [4]Sreesha C, [5]Shalini A, [6]Zeeshan Karrar
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Smart Helmet is an innovative safety system designed to improve the safety of motorcycle riders by integrating smart technologies into a conventional helmet. Road accidents involving two-wheelers often result in serious injuries due to improper helmet usage, drunk driving, and delayed emergency assistance. This project aims to develop a smart helmet that enhances rider safety by monitoring helmet usage and detecting potentially dangerous situations. The proposed system uses sensors and a microcontroller to perform various safety functions. A helmet-wearing sensor detects whether the rider is wearing the helmet, while an alcohol sensor can identify alcohol consumption before the vehicle is started. An ignition control mechanism can prevent the motorcycle from starting if the helmet is not worn or alcohol is detected. Additionally, the system can incorporate an accident detection sensor and a GPS/GSM module to identify accidents and send emergency alerts with the rider's location to predefined contacts. These features help promote responsible riding and support faster emergency response.The main objective of this project is to reduce road accident risks, encourage proper helmet usage, and improve emergency assistance for motorcycle riders. The Smart Helmet provides a practical and technology-based approach to road safety and can contribute to safer transportation.

**Keywords:** *Smart Helmet, Road Safety, Accident Detection, Alcohol Detection, Helmet Sensor, GPS, GSM, Microcontroller.*

## Smart Fire, Flame Detection and Alert System

[1]Shreekeerthi N, [2]Tanuja Y L, [3]Spoorthi M P, [4]Sakshi M, [5]Shravya V, [6]Hansika N
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Smart Fire and Flame Detection and Alert System is designed to detect fire and flame at an early stage and provide immediate alerts to prevent accidents, reduce property damage, and protect human lives. Traditional fire detection systems may have limitations in detecting fire quickly or providing timely notifications in certain environments. This project aims to develop an intelligent and reliable fire detection system using suitable sensors and a microcontroller. The proposed system uses a flame sensor and other environmental sensors, such as a temperature or smoke sensor, to monitor the surroundings continuously. When fire, flame, or abnormal temperature levels are detected, the system processes the sensor data and activates an alarm or buzzer to alert people nearby. It can also incorporate a GSM or Wi-Fi module to send real-time notifications to the user or designated authorities. The system is designed for applications in homes, offices, industries, laboratories, and other fire-prone environments. The main objective of this project is to improve fire safety through early detection, rapid alerts, and continuous monitoring. By providing a cost-effective and efficient solution, the Smart Fire and Flame Detection and Alert System can help minimize fire-related risks and support faster emergency responses.

**Keywords:** *Smart Fire Detection, Flame Sensor, Temperature Sensor, Smoke Detection, Microcontroller, Alert System, Fire Safety.*

## Smart Watering System for Indoor Plants

[1]Pallavi T, [2]Aleena Paul, [3]Hemalatha S, [4]Christo Girosh, [5]Sreenandh T, [6]Yeshas L P
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Smart Watering System for Indoor Plants is an automated solution that helps maintain the right amount of moisture in the soil without the need for regular manual watering. Many indoor plants suffer from overwatering or underwatering, which can negatively affect their growth and overall health. This system is designed to solve that problem by using a soil moisture sensor, a microcontroller such as Arduino, and a small water pump. The soil moisture sensor continuously checks the moisture level of the soil and sends the information to the microcontroller. When the soil becomes too dry and the moisture level drops below a preset value, the microcontroller automatically turns on the water pump. The pump supplies water to the plant until the required moisture level is reached. Once the soil contains enough moisture, the microcontroller switches the pump off automatically. This automated process ensures that plants receive the right amount of water at the right time, reducing the chances of damage caused by improper watering. The system also helps conserve water, saves time and effort, and promotes healthy plant growth. It can be used in homes, offices, indoor gardens, and other indoor environments. Overall, the Smart Watering System is a simple, affordable, and efficient plant-care solution. In the future, it can be enhanced with Internet of Things (IoT) technology, allowing users to monitor and control the system remotely through a mobile application.

**Keywords:** *Arduino, Automation, Indoor Plants, Internet of Things (IoT), Microcontroller, Soil Moisture Sensor, Water Conservation, Water Pump, Smart Watering System*

## River Cleaning Robot

[1]Navyashree K, [2]Bhavani, [3]Shashidhar.S, [4]Pavan D S, [5]Karthick C J, [6]Akash L Bavalatti, [7]Siripireddy Umesh
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The River Cleaning Robot is an innovative robotic system designed to remove floating waste and unwanted materials from rivers, lakes, and other water bodies. Water pollution caused by plastic bottles, polythene bags, and other floating debris has become a major environmental concern, affecting aquatic life and water quality. Traditional river cleaning methods often require manual labor, consume significant time, and may expose workers to hazardous conditions. This project aims to provide an efficient, cost-effective, and eco-friendly solution for cleaning water bodies using robotic technology. The proposed system uses a floating robotic platform equipped with a waste collection mechanism, such as a conveyor belt or collection net, to gather floating waste from the water surface. A motor-driven propulsion system enables the robot to move across the water, while a microcontroller controls its movement and cleaning operations. Depending on the design, the robot can be operated remotely or equipped with sensors for semi-autonomous navigation and obstacle detection. The collected waste is stored in a designated container and can be removed for proper disposal. The main objective of this project is to reduce water pollution, minimize human effort, and improve the efficiency of waste collection in rivers and other water bodies. The River Cleaning Robot provides a practical technological approach to environmental protection and supports the development of cleaner, healthier aquatic ecosystems.

**Keywords:** *River Cleaning Robot, Water Pollution, Waste Collection, Robotics, Microcontroller, Floating Waste, Environmental Protection.*

## Fire Detection & Alert System

[1]Vinay Kumar Maurya, [2]Raghu Jayappagouda Hulyal, [3]V. Yatin, [4]Rajat Mahapatra, [5]Akshay Gowda K S, [6]Sham H S, [7]Patil Bhagvan

Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Fire Detection and Alert System is designed to detect fire at an early stage and provide immediate alerts to help prevent accidents, minimize property damage, and protect human lives. Fire incidents in homes, industries, offices, and commercial buildings can cause serious damage if they are not detected promptly. This project aims to develop a reliable and cost-effective system that continuously monitors the surrounding environment and identifies potential fire hazards using sensors and a microcontroller. The proposed system uses a flame sensor, temperature sensor, and/or smoke sensor to detect signs of fire. When the sensor readings exceed predefined safety limits, the microcontroller processes the information and activates an alarm or buzzer to warn people nearby. The system can also incorporate a GSM or Wi-Fi module to send instant alerts to the user or designated authorities. This enables faster response and helps reduce the risk of fire spreading. The system is suitable for installation in residential buildings, offices, laboratories, warehouses, and industrial environments. The main objective of this project is to improve fire safety through continuous monitoring, early fire detection, and timely alerts. By combining sensor technology with automated warning mechanisms, the Fire Detection and Alert System provides a practical solution for enhancing safety and reducing fire-related risks.

**Keywords:** *Fire Detection, Alert System, Flame Sensor, Smoke Sensor, Temperature Sensor, Microcontroller, Fire Safety, Early Warning System.*

## Smart Irrigation System

[1]Lathika Reddy B R, [2]Manasa N, [3]Dhruva G, [4]Jayanth T, [5]Manoj B, [6]Chethan G

Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Smart Irrigation System is an automatic device designed for efficient water usage and enhanced agricultural output as a result of watering plants based on the moisture in the soil. Traditional irrigation techniques tend to waste water because of over-watering and poor scheduling and monitoring of soil moisture levels continuously. This innovation intends to build an intelligent system that would minimize manual work, save water, and contribute to the well-being of crops. The Smart Irrigation System uses a soil moisture sensor, microcontroller, water pump, and water supply. Soil moisture level is continuously measured by a sensor and then sent to the microcontroller. When soil moisture level drops below the predefined value, the pump turns on, and after reaching the required value, it switches off to prevent overwatering. Temperature sensors, rain sensors, and Internet of Things technology could be incorporated into the system.

**Keywords:** *Smart Irrigation, Soil Moisture Sensor, Automatic Watering, Microcontroller, Water Conservation, Agriculture, IoT, Sustainable Farming.*

## Automatic Rain Shield for Clothes Drying System

[1]Lakshmishree R V, [2]M Keerthana, [3]Dimpu.B.M, [4]Mahalakshmi T, [5]Angelina A, [6]Manasa M
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Automatic Rain Shield for Clothes Drying System is a computer-based automatic system designed to shield clothes against unexpected rain when drying in open spaces. The weather may change suddenly, leading to wetting of the clothes. This will lead to inconveniences for the users and increase the time taken to dry their clothes. The system aims to detect any rainfall and shield the clothes in an automated way. The system incorporates a rain sensor, microcontroller, and motorized system. During good weather conditions, the clothes remain exposed to sunlight and clean air. During rain detection, the microcontroller operates the motor to extend a rain proof cover over the clothes. Once the rain stops, and the weather favors drying of the clothes, the system will be able to automatically retract the cover, depending on the control method selected. The project seeks to reduce human efforts, prevent wetting of clothes and offer effective clothes drying system.

**Keywords:** *Automatic Rain Detection, Rain Sensor, Smart Clothes Drying, Microcontroller, Motorized Shield, Automation, Weather Protection.*

## Smart Solar Street Light System

[1]G. Bhavya, [2]V Vinesha, [3]Prem A, [4]Deeraj Kumar B, [5]Nikhil H.R, [6]Kishore Kumar P
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** The Smart Solar Street Light System is an energy-efficient and eco-friendly lighting solution designed to reduce electricity consumption by utilizing solar energy. Conventional street lighting systems depend on grid electricity and often consume power even when full illumination is not required. This project aims to develop an intelligent street lighting system that uses renewable solar energy and automatic control techniques to improve energy efficiency and reduce operating costs. The proposed system consists of a solar panel, rechargeable battery, LED street light, microcontroller, and sensors such as a Light Dependent Resistor (LDR) and a motion sensor. The solar panel converts sunlight into electrical energy during the daytime, which is stored in the battery for use at night. The LDR automatically switches the street lights on when the surrounding light level decreases and switches them off during daylight. A motion sensor can be used to adjust the brightness of the lights based on the presence of pedestrians or vehicles, helping to conserve energy when roads are empty. The system can also incorporate battery monitoring to improve performance and reliability. The main objective of this project is to provide sustainable, cost-effective, and automated street lighting while minimizing energy wastage and maintenance requirements. The Smart Solar Street Light System is suitable for roads, highways, residential areas, parking lots, and public spaces. By combining g renewable energy with intelligent automation, this project contributes to energy conservation and environmental sustainability.

**Keywords:** *Smart Solar Street Light, Solar Energy, Renewable Energy, LDR Sensor, Motion Detection, LED Lighting, Microcontroller, Energy Efficiency.*

## Smart Medicine Reminder Box

[1]S Nethravathi, [2]Rachana R, [3]Vennela.B, [4]B Priya Darshini, [5]Deekshitha. A, [6]P. Harika
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Smart Medicine Reminder Box is a healthcare solution designed to remind users about taking their medications on time and in prescribed dosages. Elderly people and chronic disease patients tend to forget about taking medicines on time. This can impair the effectiveness of the treatment. The system incorporates a microcontroller, RTC module, buzzer, LEDs and different medicine compartments. Users can set their medication schedule, while the system will measure time all the time. If the time for taking medication comes, then the buzzer starts ringing and the LED starts blinking. There are additional functions that include time and schedule on LCD display, indication of compartments and messages to caregivers.

**Keywords:** *Smart Medicine Reminder Box, Medication Management, Microcontroller, RTC Module, Buzzer Alert, LCD Display, Healthcare Automation, Medication Adherence*.

## Probability Distribution Transformation Based on Diffusion Models

[1]J. Raji, [2]P Harika, [3]Chitrashree S, [4]Dimpu B. M, [5]Mahalakashmi
Dept. Artificial Intelligence and Machine Learning, T. John Institute of Technology, Bengaluru-83

**Abstract:** Diffusion-based generative artificial intelligence (AI) models have emerged as a powerful framework for generating high-quality images, audio, video, and other forms of data. Their performance is fundamentally connected to probability distributions, because generation is formulated as a sequencstatise of stochastic transformations that gradually remove noise from a random sample. This paper presents a mathematical study of probability distributions underlying diffusion-based generative AI, with particular emphasis on Gaussian noise, conditional distributions, forward diffusion, and reverse denoising processes. The methodology involves formulating the forward diffusion process as a Markov chain in which controlled Gaussian noise is progressively added to an initial data distribution. The resulting transition probabilities and their relationship to the standard normal distribution are analysed mathematically. The reverse process is then studied as a learned probabilistic transformation, where a neural network estimates the noise or score required to recover samples from the original data distribution. Key probability concepts, including expectation, variance, conditional probability, likelihood, and Kullback–Leibler divergence, are used to explain the objective functions commonly associated with diffusion models. The study also considers how the noise schedule influences the distribution of intermediate states and the quality of generated samples. The analysis indicates that probability distributions provide the mathematical foundation for both the corruption and reconstruction stages of diffusion models. Gaussian assumptions make the forward process analytically tractable, while learned reverse distributions enable complex data structures to be generated from simple random noise. The study demonstrates that changes in the noise schedule and the accuracy of the learned reverse distribution can directly affect sample quality and generation stability.

**Keywords:** *Diffusion Models, Generative Artificial Intelligence, Probability Distributions, Gaussian Noise, Markov Chain, Reverse Denoising, Conditional Probability, Kullback–Leibler Divergence*.

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tjitprincipal@tjohngroup.com

+91-8049030512

**Address:** Gottigere, Bannerghatta Road,
Bangalore, Karnataka, India - 560 083

www.tjohncollege.com

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